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# Dynamics Within Latent Chain-of-Thought: An Empirical Study of Causal Structure

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## Abstract

Latent or continuous chain-of-thought methods replace explicit textual rationales with a number of internal latent steps, but these intermediate computations are difficult to evaluate beyond correlation-based probes. In this paper, we view latent chain-of-thought as a manipulable causal process in representation space by modeling latent steps as variables in a structural causal model (SCM) and analyzing their effects through step-wise do-interventions. We study two representative paradigms (i.e., Coconut and CODI) on both mathematical and general reasoning tasks to investigate three key questions: (1) which steps are causally necessary for correctness and when answers become decodable early; (2) how influence propagates across steps and how this structure compares to explicit CoT; and (3) whether intermediate trajectories retain competing answer modes and how output-level commitment differs from representational commitment across steps. We find that latent-step budgets behave less like homogeneous extra depth and more like staged functionality with non-local routing, and we identify a persistent gap between early output bias and late representational commitment. These results motivate mode-conditional and stability-aware analyses, together with corresponding training/decoding objectives, as more reliable tools for interpreting and improving latent reasoning systems. Code is available at <https://github.com/J1mL1/causal-latent-cot>.

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## 1. Introduction

Large language models (LLMs) have achieved strong performance on mathematical problem solving and logical question answering (Cobbe et al., 2021; Geva et al., 2021). A widely adopted technique is Chain-of-Thought (CoT) prompting, which improves accuracy by eliciting intermediate reasoning steps in natural language (Wei et al., 2022). Despite its empirical effectiveness, explicit CoT incurs substantial decoding cost, often produces verbose outputs, and may contain post-hoc rationalizations that do not faithfully reflect the computations driving model predictions (Pruthi et al., 2020; Turpin et al., 2023). These limitations motivate a shift from reasoning in tokens to reasoning in representations.

Recent work explores latent/continuous CoT, where multi-step inference is carried out in continuous hidden representations rather than long textual explanations (Hao et al., 2025; Shen et al., 2025; Zhang et al., 2025b; Xu et al., 2025; Wei et al., 2026; Gozeten et al., 2026). This paradigm promises a higher-bandwidth internal workspace and reduced decoding overhead, but it faces a fundamental interpretability challenge: 1) intermediate computations are no longer exposed as discrete, human-editable steps; 2) reasoning-relevant information is often distributed across latent dimensions and iterative steps. Hence, traditional analytical methods—such as step editing or ablation—cannot be directly applied to implicit CoT, leaving the causal and mechanistic analysis of implicit CoT underexplored.

Motivated by this research gap, we conceptualize latent CoT as a **causal system** (Pearl, 2000; Yao et al., 2021) evolving over latent-step variables and evaluate it with intervention-based causal analysis (Robins & Greenland, 1992; Imai et al., 2010). Specifically, we treat model-defined intermediate latent states as the unit of a reasoning “step” and perform step-wise do-interventions (Singh et al., 2024; Kaddour et al., 2025) that modify an intermediate state while keeping downstream computation unchanged. We then quantify the causal sensitivity of the model’s output to these interventions. Furthermore, we aggregate step-to-step influences to construct a directed acyclic graph (Thulasiraman & Swamy, 1992) of information flow, derived from intervention-induced shifts in teacher-forced

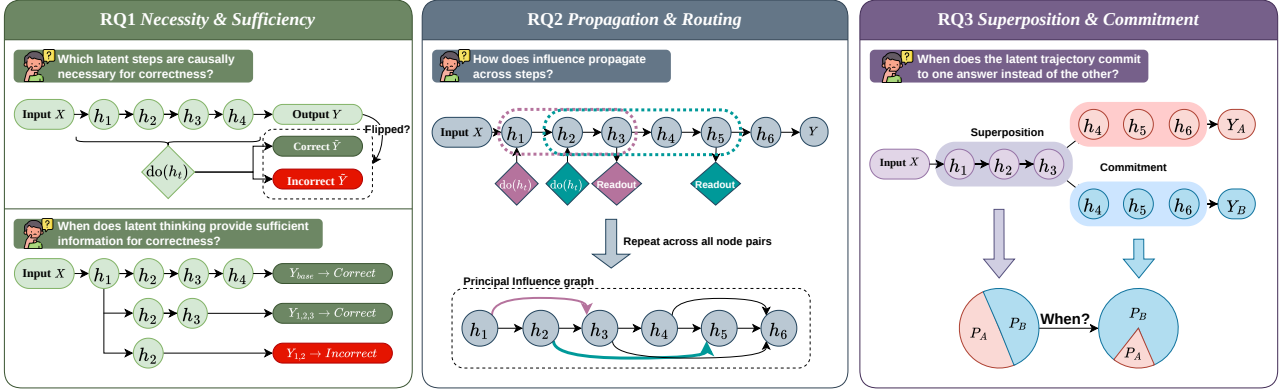


Figure 1. Overview of step-centric research questions for latent CoT. RQ1 tests step necessity and early decodability; RQ2 characterizes step-to-step influence propagation; RQ3 probes trajectory-level superposition and commitment across rollouts.

output distributions. This graph captures how information propagates through the latent reasoning process. Figure 1 provides a high-level roadmap of the three questions we study: **(RQ1)** which latent steps are causally necessary for correctness and when the answer becomes decodable, **(RQ2)** how influence propagates across steps, and **(RQ3)** whether intermediate trajectories retain competing hypotheses and how commitment evolves across steps.

We instantiate this causal evaluation framework on two representative latent-reasoning paradigms, Coconut (Hao et al., 2025) and CODI (Shen et al., 2025), across both mathematical and general reasoning tasks. Empirically, we find a phenomenon–mechanism–nature pattern: **(phenomenon)** causal leverage is highly heterogeneous across latent steps, with a small subset exerting outsized influence; **(mechanism)** step-to-step effects are often non-local, indicating routed propagation rather than purely chain-like transmission; and **(nature)** output-level preference can emerge earlier than representational consolidation, revealing a persistent gap between early bias and later commitment.

Our main contributions are threefold: **(1) the first causal, step-resolved evaluation view of latent CoT** that distinguishes when a solution becomes available from which steps remain causally necessary; **(2) an operator- and readout-conditioned influence analysis** that recovers dominant propagation routes while avoiding sparsity overclaims; and **(3) mode-conditional evidence that links latent-step budgets to practical design implications:** early “decision signals” need not imply early commitment, so improving latent reasoning likely requires shaping routing/commitment rather than simply adding more steps.

## 2. Related Work

**Latent and continuous chain-of-thought reasoning.** To avoid the cost and potential unfaithfulness of explicit CoT, recent methods move reasoning into continuous

representations (Chen et al., 2025; Zhu et al., 2025b). A first family performs depth-iterative latent reasoning, where the model recurrently updates a hidden state or continuous “thought token” before decoding, as in Coconut (Hao et al., 2025) and CODI (Shen et al., 2025). Related approaches incorporate additional supervision or alignment to stabilize latent reasoning (Wei et al., 2026; He et al., 2025), while hybrid methods mix latent and textual steps to balance efficiency and interpretability (Su et al., 2025). A second family scales test-time computation through recurrent-depth paradigms and parallel continuous updates (Geiping et al., 2025; Wu et al., 2025; Zhu et al., 2025c). Most literature emphasizes accuracy and efficiency; our work focuses on internal causal organization within latent reasoning and how it relates to explicit CoT.

**Large reasoning models and reinforcement learning for reasoning.** Recent work also improves LLM reasoning through reinforcement learning, selective supervision, uncertainty estimation, and long-horizon inference. Representative directions include RL-based training and surveys for large reasoning models (Zhang et al., 2025a), fine-grained reward signals for test-time reinforcement learning (Wang et al., 2025), critical-token fine-tuning (Ruan et al., 2025), internal-belief-based uncertainty estimation (Xiao et al., 2025), and infinite-horizon reasoning via reinforcement learning (Yan et al., 2026).

**Chain-of-thought faithfulness and causal tests.** A growing literature questions whether model-provided explanations faithfully reflect the computations that determine final predictions. Models can generate convincing yet unfaithful artifacts, including attention explanations, CoT rationales, and structured parses (Pruthi et al., 2020; Turpin et al., 2023; Bai et al., 2025). Prior work assesses faithfulness through context- or rationale-level interventions, such as deleting, shuffling, or editing rationales, and measures changes in downstream answers (Wang et al., 2023a; Yang et al., 2023; Yu et al., 2025). Complementary

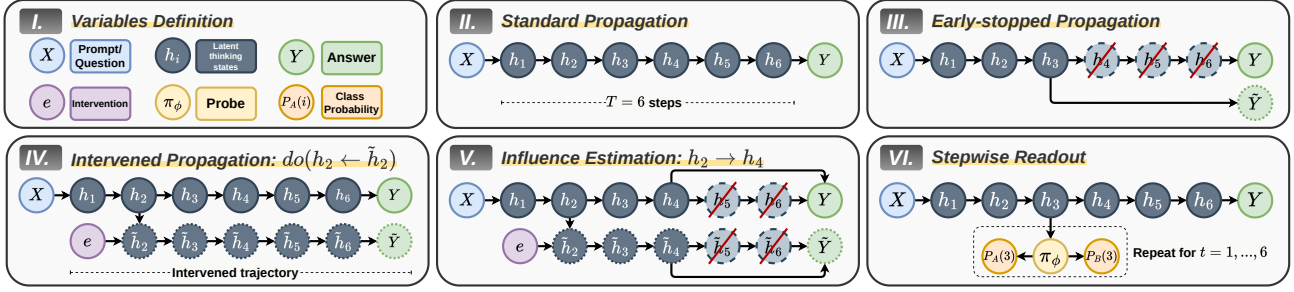


Figure 2. **Intervention-based protocol for latent CoT as a causal system.** (I) **Variables.** Input  $X$  induces a latent trajectory  $\{h_t\}_{t=1}^T$  and output  $Y$ ; an intervention operator implements step-wise  $\text{do}(h_t \leftarrow \tilde{h}_t)$ ; a readout maps hidden states to answer support. (II) **Standard Propagation.** Unperturbed dynamics from  $X$  through steps to  $Y$ . (III) **Early-stopped Propagation.** We truncate latent computation after step  $k$  and decode from  $h_k$  to test when correctness becomes decodable (RQ1). (IV) **Intervened Propagation.** We replace a step’s state while keeping downstream computation intact, yielding an intervened outcome  $\tilde{Y}$  (RQ1). (V) **Influence Estimation.** Combining a step- $t$  intervention with an early readout at step  $s$  yields directed propagation strengths  $W_{t,s}$  summarized as an empirical influence structure (RQ2). (VI) **Stepwise Readouts.** We read out the answer competition from  $h_t$  (e.g., teacher forcing or a probe) to characterize superposition and commitment (RQ3).

parameter-level interventions, such as unlearning reasoning steps from model weights, provide another causal probe of whether a step is internally represented and used (Tutek et al., 2025).

**Causal and mechanistic analysis of internal representations.** Causal perspectives view hidden activations as mediators that can be intervened on to test their functional role in model behavior (Pearl, 2000; Feder et al., 2022; Schölkopf et al., 2021). Causal mediation analysis and mechanistic interventions have been used to distinguish behaviorally relevant internal components from merely correlated representations (Vig et al., 2020; Finlayson et al., 2021; Stolfo et al., 2023; Serrano & Smith, 2019; Chrysostomou & Aletras, 2021; Elhage et al., 2022; Nanda et al., 2023; Meng et al., 2022; Wang et al., 2023b; Conmy et al., 2023). Recent work further shows that models can internalize reasoning strategies while concealing them from surface text under process supervision, highlighting the gap between latent computation and explicit explanation (McCarthy et al., 2025). These toolkits motivate our step-level interventions on latent reasoning states.

### 3. Evaluation Framework: Latent CoT as a Causal System

#### 3.1. Scope and Causal Queries

We view latent CoT as a manipulable causal process evolving in continuous representation space. Specifically, we treat intermediate latent reasoning steps as variables in a structural causal model (SCM) (Pearl, 2000; Yao et al., 2021), which allows us to pose intervention-based causal queries and compute reproducible effect estimates under a fixed intervention protocol (Robins & Greenland, 1992; Imai et al., 2010; Vig et al., 2020; Mueller et al., 2026).

Our analysis provides a unified intervention + readout protocol (Figure 2) and uses it to answer three step-centric causal questions. **(RQ1: necessity and sufficiency)** asks whether individual latent steps are behaviorally necessary, and whether the full latent budget is required to make the correct answer decodable. **(RQ2: propagation and routing)** asks how a perturbation at an upstream step propagates to downstream computation, summarized as a step-to-step influence matrix and visualized as an empirical influence graph. **(RQ3: superposition and commitment)** asks whether intermediate trajectories retain competing answer hypotheses and how output-level commitment relates to representational commitment across steps.

#### 3.2. Causal Variables, Minimal SCM, and Latent-step Interface

For an input problem  $x$ , we model latent reasoning as a sequence of continuous causal variables  $H_{1:T} = (H_1, \dots, H_T)$  with  $H_t \in \mathbb{R}^d$ , where each  $H_t$  corresponds to the model’s internal latent state at reasoning step  $t$ . We denote the task-level output by  $Y$  (e.g., the final predicted answer/label), and treat it as a random variable whose conditional distribution is induced by the model’s decoder given the latent trajectory. A minimal SCM consistent with this computation is

$$H_t = f_t(H_{<t}, x, \epsilon_t; \theta), \quad t = 1, \dots, T, \quad (1)$$

$$Y = g(H_{1:T}, x, \epsilon_y; \theta), \quad (2)$$

where  $f_t$  and  $g$  are the model’s fixed transition and decoding mechanisms, and  $\epsilon_t, \epsilon_y$  capture stochasticity.

In our experiments, we intervene on the intermediate states produced by a latent thinking rollout. We denote realized latent states by lowercase  $h_t$  and write

$$h_{1:T} \sim p_\theta(H_{1:T} | x), \quad (3)$$

where  $h_t \in \mathbb{R}^d$  instantiates the random variable  $H_t$  at step  $t$  under the model dynamics. Latent-reasoning models expose these variables through a fixed-length sequence of hidden states  $h_{1:T} = (h_1, \dots, h_T)$ , where  $h_t$  is the last-layer hidden representation associated with the  $t$ -th latent step (e.g., a continuous “thought token” in COCONUT or the designated reasoning position in CODI) and is used as the step- $t$  reasoning input embedding.

**Definition 3.1** (Single-step latent do-intervention). Given input  $x$  and a baseline rollout realizing  $h_{1:T}$ , the intervention  $\text{do}(h_t \leftarrow \tilde{h}_t)$  replaces the step- $t$  latent state by  $\tilde{h}_t$  and recomputes all downstream states using the same transition mechanism, yielding a counterfactual trajectory  $\tilde{h}_{1:T}$ . Formally, let  $\tilde{h}_{<t} = h_{<t}$  and  $\tilde{h}_t$  be the overwritten state; for  $t' > t$  we set

$$\tilde{h}_{t'} := f_{t'}(\tilde{h}_{<t'}, x, \tilde{\epsilon}_{t'}; \theta), \quad (4)$$

where  $\tilde{\epsilon}_{t'}$  matches the baseline randomness when applicable. The corresponding counterfactual output is obtained by the same readout mechanism,

$$\tilde{y} = g(\tilde{h}_{1:T}, x, \tilde{\epsilon}_y; \theta). \quad (5)$$

Unless otherwise stated, we use deterministic rollouts whenever possible; otherwise, we control randomness (e.g., fixed seeds) and isolate propagation effects via teacher-forced readouts (Figure 2(V)) to reduce sampling noise.

### 3.3. Paradigms of Latent-reasoning Models

We instantiate our framework on two latent-reasoning paradigms that differ in how they realize latent steps.

**Coconut** (Hao et al., 2025) employs an explicit latent mode: it uses the final hidden state as a continuous reasoning token, feeding it back as the input for the next step, rather than decoding a discrete token.

**CODI** (Shen et al., 2025) compresses discrete CoT into continuous space via self-distillation: a continuous-CoT student is trained to both produce the correct answer and align its hidden states, at specific reasoning steps, with those of a discrete-CoT teacher, encouraging the latent trajectory to inherit stepwise structure.

### 3.4. Models and Data

**Models.** We experiment with both CODI and COCONUT on multiple backbones. We use official CODI checkpoints for GPT-2 (Radford et al., 2019) and Llama3-1B (Grattafiori et al., 2024), and reproduce it on Qwen3-4B-Instruct (Yang et al., 2025). Coconut models are reproduced across the same three backbones (GPT-2, Llama3-1B, Qwen3-4B-Instruct). Implementation details are in Appendix A. Additional checks on Sim-CoT (Wei et al., 2026) and

larger/alternative COCONUT backbones are reported in Appendix D and Appendix E, respectively.

**Datasets.** We train and evaluate on CoT-augmented datasets that follow previous latent-reasoning methods. For GSM8K, we train on GSM8K-Aug (Deng et al., 2023) and evaluate on the original GSM8K test set (Cobbe et al., 2021). For CommonsenseQA, we use the CoT-augmented training set released by CODI (Shen et al., 2025) and evaluate on the original CommonsenseQA test set (Talmor et al., 2019). Dataset details are provided in Appendix B.

## 4. RQ1: Step-wise Necessity and Sufficiency

Latent reasoning replaces long textual rationales with a fixed-length sequence of intermediate hidden states, enabling step-wise manipulation under the intervention-based protocol as introduced in Sec. 3 (Figure 2). RQ1 asks two causal questions aligned with this protocol: (i) **step-wise necessity**, i.e., whether the final decoded decision depends on a particular intermediate state, and (ii) **latent-budget sufficiency**, i.e., how many latent steps are required before the correct answer becomes decodable. To address (i), we treat each realized latent state  $h_t$  as an intervenable object and run paired rollouts that differ only by a single-step do-intervention (Def. 3.1) on  $h_t$ , quantifying its impact on the final decoded answer. For (ii), we perform early-stop decoding by truncating latent computation after step  $k$  and decoding directly from  $h_k$  (Figure 2(III)) to determine when correctness first becomes readable from the latent trajectory.

### 4.1. Step-wise Interventions

#### 4.1.1. EXPERIMENT SETTING

We evaluate step-wise necessity using the step-wise do-intervention in Def. 3.1. For each example, we run a baseline rollout and an intervened rollout that modifies exactly one latent state  $h_t$  while keeping all other components unchanged, including the input  $x$ , parameters  $\theta$ , and the downstream transition and readout mechanisms. Following Appendix A.2, we adopt the **zero intervention** for its simplicity and consistency across different models:

$$\text{do}(h_t \leftarrow \tilde{h}_t), \quad \tilde{h}_t = \mathbf{0}. \quad (6)$$

We then decode the final prediction  $\tilde{y}^{(t)}$  and compute the **flip rate**  $\text{Flip}(t)$ , defined as the fraction of examples for which  $\tilde{y}^{(t)}$  differs from the baseline output (Serrano & Smith, 2019; Chrysostomou & Aletras, 2021). This metric directly measures decision-level dependence on the latent computation at step  $t$  under a fixed intervention operator.

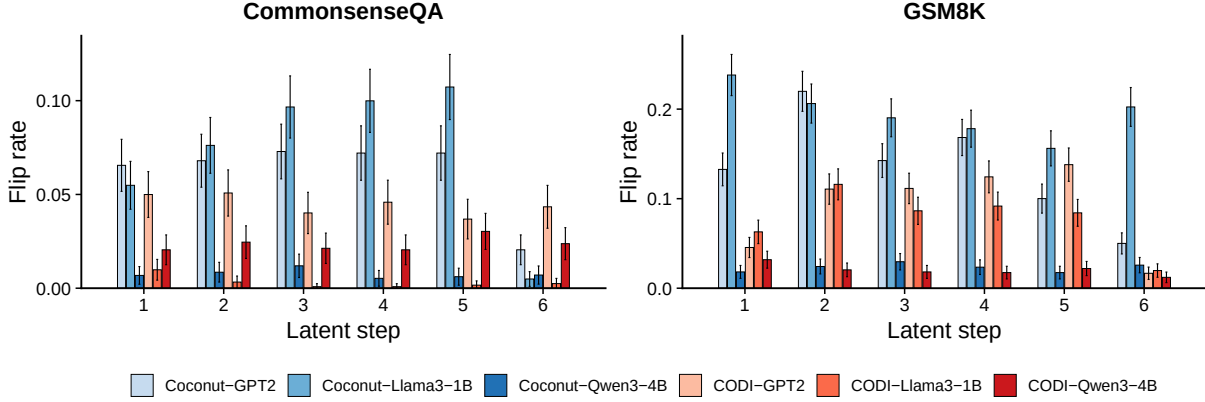


Figure 3. **Step-wise necessity measured by decision instability.** We intervene at a single latent step  $t \in \{1, \dots, 6\}$  by zeroing its state,  $\text{do}(h_t := 0)$ , and then decode the final answer. We report the flip rate  $\text{Flip}(t)$ , i.e., the fraction of examples whose decoded prediction changes relative to the baseline, on CommonsenseQA (left) and GSM8K (right). Error bars indicate estimation uncertainty.

#### 4.1.2. FINDINGS

**Single-step ablation produces clear, step-specific flip patterns (not uniformly sensitive).** Within the same model and dataset,  $\text{Flip}(t)$  changes noticeably with the intervened step  $t$  (Figure 3), with several settings exhibiting mid-step peaks rather than a flat or monotone pattern. This indicates that different latent steps contribute differently to the final decision, and that single-step removal can selectively disrupt the decision more at some steps than others.

**Arithmetic exhibits substantially higher decision volatility than commonsense.** Flip rates on GSM8K are markedly higher than on CommonsenseQA for the same intervention protocol, with several backbones reaching flip rates around 0.1–0.2 or higher on GSM8K, while CommonsenseQA remains mostly below  $\sim 0.1$  (Figure 3). This gap is visible across both Coconut and CODI variants.

**Coconut shows larger flips than CODI under matched backbones, while stronger backbones suppress flips.** Under the same backbone, Coconut variants generally yield higher flip rates than CODI, especially on GSM8K (Figure 3). Moreover, while stronger backbones substantially reduce flip rates across both paradigms, the flipping profile remains step-dependent—even when absolute rates are low.

#### 4.1.3. IMPLICATIONS

These intervention results indicate that causal leverage is unevenly distributed across the latent trajectory. The effect of replacing a single latent state depends on where that state occurs and on the model–task setting in which it is used, rather than being an invariant property of the latent-step index. Thus, step-wise intervention identifies position-specific decision sensitivity within a fixed rollout, supporting a view of latent steps as differentiated components rather than uniformly interchangeable states.

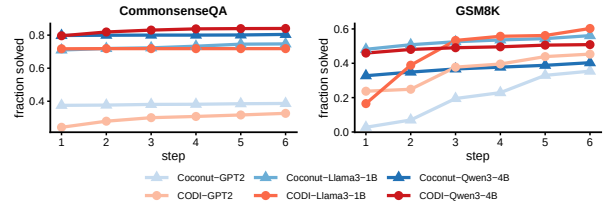


Figure 4. **Early-stop decoding reveals when correctness becomes decodable.** We report the cumulative solved fraction  $S(k) = \mathbb{P}(k_i \leq k)$  under early-stop decoding on CommonsenseQA (left) and GSM8K (right), where  $k_i$  is the earliest step at which the correct answer becomes decodable (Def. 4.1).

## 4.2. Early-stop Decoding

### 4.2.1. EXPERIMENT SETTING

**Definition 4.1** (Earliest correct step under early-stop decoding). For each example  $i$ , let  $\hat{y}_i^{(\leq k)}$  be the decoded answer when we truncate latent computation after step  $k$  and decode from  $h_{i,k}$ . We define the earliest step at which the correct answer becomes decodable as

$$k_i = \min\{k \in \{1, \dots, T\} : \hat{y}_i^{(\leq k)} = y_i^*\}, \quad (7)$$

and set  $k_i = \infty$  if no early-stop decode is correct for  $k \leq T$ .

**Definition 4.2** (Cumulative solved fraction). Let  $k_i$  be defined in Def. 4.1. We define the cumulative solved fraction

$$S(k) = \frac{1}{N} \sum_{i=1}^N \mathbf{1}\{k_i \leq k\}. \quad (8)$$

### 4.2.2. FINDINGS

**Early-stop curves differ across datasets.** On CommonsenseQA,  $S(k)$  typically rises rapidly within the first few steps and then saturates (Figure 4, left), suggesting most solvable instances become decodable within few steps. In contrast, on GSM8K  $S(k)$  often continues to increase

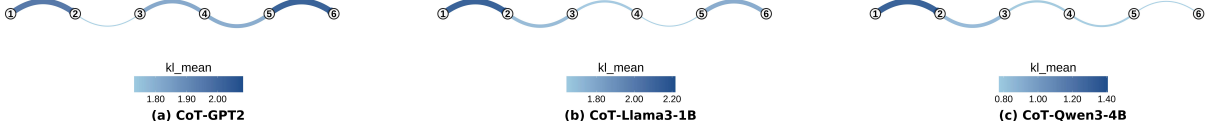


Figure 5. **Explicit CoT principal influence graphs (GSM8K; CoT-SFT baselines)**. Nodes denote the first  $T=6$  segmented CoT steps. Edge  $t \rightarrow s$  indicates propagation strength  $W_{t,s}$  from Eq. 10 (teacher-forced KL shift on the gold answer under a single-step intervention at  $t$  and readout at  $s$ ). For readability, we show only top-1 outgoing edges after thresholding at  $\alpha=0.1 \cdot \max(W)$ .

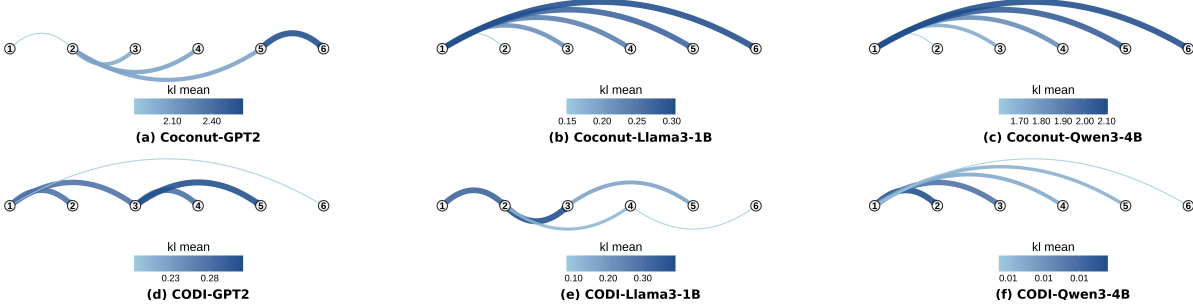


Figure 6. **Latent principal influence graphs (GSM8K; COCONUT/CODI)**. Nodes are latent steps  $t \in \{1, \dots, 6\}$ . Edge weights follow Eq. 10 under single-step interventions, rendered with the same sparsification protocol as Figure 5.

toward later steps (Figure 4, right), with several settings showing gains up to  $k=6$ , indicating that additional latent computation can expand the set of instances for which the correct answer becomes decodable.

#### Backbone strength shapes when correctness becomes decodable, while paradigm-level similarity is weak.

Across both datasets, stronger backbones have higher  $S(1)$  and earlier saturation of  $S(k)$ , whereas weaker backbones improve more gradually with  $k$  (Figure 4). Meanwhile, the step-wise profiles do not cluster by training paradigm: Coconut and CODI do not consistently exhibit similar curve shapes within each paradigm across backbones, suggesting that “when correctness becomes decodable” is not a stable paradigm-level signature in this experiment.

#### 4.2.3. IMPLICATIONS

Early-stop decoding complements intervention analysis by measuring answer availability rather than step necessity. The growth and saturation of  $S(k)$  show that additional latent computation can expose correct-answer information for more instances, but the required budget depends on task difficulty and backbone capacity. Thus, latent steps can be causally differentiated while still accumulating useful evidence across the trajectory.

## 5. RQ2: Information Flow and Stepwise Influence Structure

RQ1 establishes that intervening on a single latent step can change the final decision, but it does not reveal where this perturbation propagates along the reasoning

trajectory. In RQ2, we compare step-to-step propagation of explicit CoT and latent CoT using an influence matrix  $W$  (Eq. 10) and its principal influence graph rendering for readability. We couple a single-step intervention at step  $t$  with an early readout at a downstream step  $s$ , and measure how strongly the intervention changes the teacher-forced output distribution for the gold answer; see Figure 2 (V). Unless otherwise stated, we report GSM8K in the main text; CommonsenseQA results are provided in Appendix C.2. Because the resulting graph is protocol-conditioned, we additionally test alternative intervention operators and an alternative readout protocol in Appendix C.4 and Appendix C.5.

### 5.1. Experiment Setting

**Nodes.** For latent-reasoning models, each node corresponds to a latent step  $t \in \{1, \dots, T\}$  with state  $h_t$  (Sec. 3). For explicit CoT baselines (CoT-SFT), we segment the generated rationale into at most  $T=6$  steps and represent step  $t$  by the last-layer hidden state at the final token of that segment.<sup>1</sup> This yields matched step-level trajectories for latent and explicit reasoning.

**Edges via intervention + early decoding.** To probe directed propagation from step  $t$  to a downstream step  $s > t$ , we run (i) a baseline rollout and (ii) an intervened rollout that modifies exactly one step state while keeping downstream computation unchanged. We then *decode at step  $s$*  using teacher forcing to obtain output distributions  $p_{\text{base}}^{(s)}(\cdot)$  and

<sup>1</sup>If an example has fewer than  $T$  CoT steps, the remaining nodes are absent.

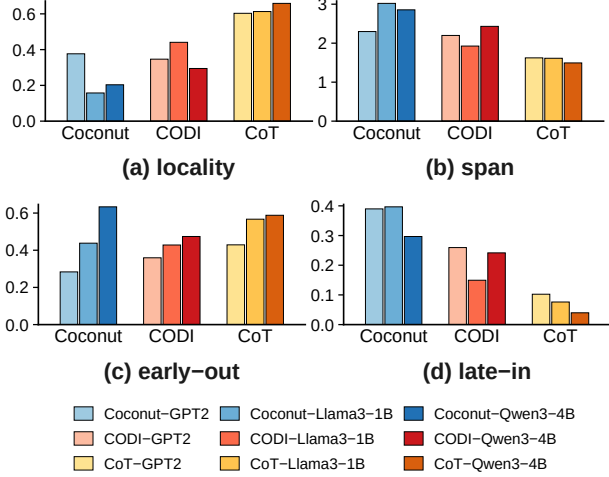


Figure 7. **Structural summaries of influence matrices (GSM8K).** We report locality, span, early-out, and late-in computed on dense normalized  $W$  (Appendix C.3).

$p_{\text{do}(t)}^{(s)}(\cdot)$ , and define the example-level propagation strength as a position-averaged KL shift:

$$\text{KL}_{t \rightarrow s}^{(i)} = \frac{1}{|y_i^*|} \sum_{u=1}^{|y_i^*|} \text{KL} \left( p_{\text{base}}^{(s)}(\cdot | y_{i, < u}^*) \parallel p_{\text{do}(t)}^{(s)}(\cdot | y_{i, < u}^*) \right). \quad (9)$$

Aggregating over evaluation examples yields the influence matrix  $W \in \mathbb{R}^{T \times T}$ :

$$W_{t,s} = \mathbb{E}_i [\text{KL}_{t \rightarrow s}^{(i)}], \quad t < s, \quad (10)$$

an operator-specific empirical influence structure under a fixed intervention and readout protocol, rather than a uniquely identified “true” causal graph.

**Principal influence graph rendering.** For compact comparison across models, we visualize  $W$  as a sparsified principal influence graph: we drop edges below  $\alpha \cdot \max(W)$  with  $\alpha=0.1$  and retain only the top-1 outgoing edge per node. Edge thickness scales with  $W_{t,s}$ , and dense heatmaps are provided in Appendix C.1.

**Structural summaries.** To quantify the heatmap-level patterns beyond visual inspection, we compute four normalized structure metrics on the dense  $W$  (definitions in Appendix C.3): **locality** (mass near the diagonal), **span** (expected hop distance), **early-out** (influence originating from early steps), and **late-in** (influence terminating at late steps). We normalize  $W$  to remove scale effects across backbones before computing these metrics.

## 5.2. Findings

**Explicit CoT influence graphs remain near-chain across backbones.** CoT-SFT exhibits a consistently sequential topology: the dominant edges follow adjacent transitions,

with only minor deviations across backbones (Figure 5). This stability is also reflected in the structure summaries on GSM8K (Figure 7): CoT-SFT has uniformly high locality (all  $\geq 0.6$ ) with low span, matching the intuition that textual steps induce predominantly local dependencies.

**Latent influence graphs are dominated by skip connections, revealing non-chain propagation.** Latent graphs contain substantially more skip connections than explicit CoT (Figure 6), indicating that influence often bypasses intermediate steps rather than accumulating strictly along a local chain. This is also captured by the structure summaries on GSM8K (Figure 7): latent models are markedly less local and have larger spans than CoT-SFT, and they place substantially more normalized influence into late-step targets (late-in). Within this skip-dominant regime, COCONUT tends to exhibit more pronounced early $\rightarrow$ late routing (often connecting early steps directly to the final step), while CODI departs from a strict chain but is generally less dominated by early $\rightarrow$ final shortcuts and shows greater variation across backbones (Figure 6). Despite this difference between the two paradigms, the skip connections, together with the non-uniform sensitivity observed in Figure 3, indicate functional differentiation across nodes in the latent trajectory.

## 5.3. Implications

These influence patterns suggest that latent reasoning does not simply preserve the causal flow of the explicit CoT trajectories from which it is derived. Although COCONUT and CODI both obtain latent computation by replacing or compressing explicit reasoning steps into continuous states, their step-to-step influence structures depart from the near-chain topology observed in explicit CoT. Under the same intervention and readout protocol, explicit CoT remains close to local sequential propagation, whereas latent CoT exposes longer-range routes that often terminate at later readouts. This implies that latentization changes not only the surface format of reasoning, but also its internal routing structure: computational adjacency in latent space need not mirror textual adjacency in natural-language rationales.

## 6. RQ3: Superposition and Commitment in Latent Dynamics

RQ1 demonstrates that removing a single latent step can alter the final decision, while RQ2 reveals step-to-step influence is non-local: early computation can directly affect multiple later steps, bypassing adjacent ones. A remaining ambiguity is whether such non-locality reflects (i) *early commitment* to one answer mode, which is then propagated, or (ii) *sustained competition* among multiple hypotheses within the latent trajectory. Prior work offers contrasting views: some argue continuous “soft thinking” remains

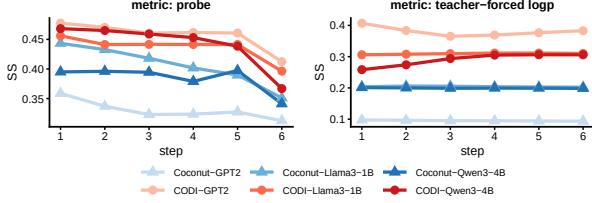


Figure 8. **RQ3: Step-wise superposition on StrategyQA.** Left: probe readout; right: teacher-forced log-probability readout. The superposition score  $SS$  measures how much support is simultaneously retained for Yes and No at each latent step.

effectively greedy and single-threaded (Wu et al., 2026), while others suggest latent reasoning may retain competing hypotheses in shared subspaces (Zhu et al., 2025a).

RQ3 therefore asks a trajectory-level question: **when stochastic rollouts of the same prompt lead to different final answers, do intermediate steps exhibit superposition, and how does this evolve across steps?** We instantiate this analysis on **StrategyQA** (Geva et al., 2021), whose binary label space (Yes/No) clearly defines two modes and allows direct probability-based readout at each step. For GSM8K, whose open-ended numeric answers make two-mode filtering substantially sparser, we provide a supporting analysis in Appendix F.3.

### 6.1. Experiment Setting

**Two-mode prompts from stochastic rollouts.** For each prompt  $x$ , we enable stochastic decoding and sample  $K$  rollouts. Each rollout produces a latent trajectory  $\{h_t^{(k)}\}_{t=1}^T$  and a final answer  $\hat{y}^{(k)} \in \{\text{Yes}, \text{No}\}$ . We retain prompts whose rollouts contain both answers, and partition rollouts into  $\mathcal{C}_Y$  and  $\mathcal{C}_N$  accordingly. Filtering thresholds and the filtering strategy are reported in Appendix F.1.

**Intermediate-step readouts (teacher-forced vs. probe).** At each latent step  $t$ , we quantify the model’s relative support for Yes vs. No using two readouts: (i) **teacher-forced** scoring under a fixed answer template, and (ii) a lightweight **probe** trained on frozen latent states to predict the rollout mode. Both yield a step-wise binary distribution over the two answer modes, denoted by  $p_Y(t)$  and  $p_N(t)$ . Templates and probing details are provided in Appendix F.2.

**Superposition score.** Given  $p_Y(t), p_N(t)$ , we define the step-wise superposition score

$$SS(t) = \min(p_Y(t), p_N(t)), \quad (11)$$

which is high when both answers retain non-trivial support and low when one answer dominates.

### 6.2. Findings

**Teacher-forced readout suggests early output-level commitment.** In Figure 8 (right),  $SS(t)$  is uniformly low and varies only modestly across steps, indicating that the model’s answer distribution becomes skewed toward either Yes or No early in the latent budget. A minimal implication is that the non-local edges in RQ2 can be compatible with early mode selection followed by propagation, without requiring prolonged output-level ambiguity (cf. the non-local principal influence graphs in Figure 6).

**Probe readout reveals more sustained competition and a late collapse.** In Figure 8 (left), the probe panel shows substantially higher  $SS(t)$  throughout the trajectory, with a clear drop at the final step, implying that intermediate states can retain decodable support for the alternative mode even when teacher forcing appears committed. This readout gap highlights readout dependence: intermediate representations may remain “multi-mode available” beyond what is expressed by the default answer distribution.

**Paradigm-level separation aligns with RQ2’s structure differences.** Under the probe readout, CODI variants maintain higher superposition scores than COCONUT variants across latent steps (Figure 8, left). This separation is consistent with RQ2: COCONUT more often exhibits dominant early-to-late principal edges, whereas CODI shows less extreme long-range routing (Figures 6 and 7).

### 6.3. Implications

These readouts distinguish output-level commitment from representational commitment. Teacher-forced scoring indicates that the default answer distribution can become biased toward one mode early, while probe readout shows that alternative modes can remain accessible in the intermediate states. Thus, commitment in latent reasoning is not a single collapse event: output preference may emerge before the latent trajectory has fully eliminated competing representational modes.

## 7. Discussion

Our experiments connect step-wise necessity and early decodability (RQ1), directed propagation (RQ2), and trajectory-level mode dynamics (RQ3). Across COCONUT and CODI, the results suggest that a fixed latent budget behaves less like homogeneous extra depth than a structured interface: steps have unequal causal leverage, influence can route non-locally, and output-level commitment need not coincide with representational commitment.

**Latent steps are causally functional, with heterogeneous leverage.** RQ1 shows that latent computation is not merely incidental to decoding: single-step interventions can alter

the final decision, but this causal dependence is unevenly distributed across the latent budget. This pattern suggests an implicit *division of labor*, where some steps act as high-leverage intervention sites while others contribute more conditionally, depending on the input or reasoning mode. Rather than repeatedly refining a homogeneous state, latent reasoning appears to introduce step-specific updates whose downstream effects are later amplified, transformed, or gated. The different leverage profiles of COCONUT and CODI further suggest that the training paradigm shapes where decision-relevant dependence concentrates.

**Principal influence structure reveals functional routes of latent computation.** Step-wise necessity profiles alone do not identify *how* a perturbation travels through the remaining latent trajectory. The principal influence graphs from RQ2 add this missing structure by highlighting dominant directed routes through which an intervention at step  $t$  manifests at later readouts. A recurring implication is that latent computation can be organized around a small number of effective long-range paths, where early steps shape later representations without requiring strong adjacent mediation at every intermediate step. This structural view helps reconcile cases where a step exhibits modest direct necessity but still participates in a high-influence route: its role may be to shape downstream states that only become consequential after later consolidation.

**Superposition and commitment: shared computation versus collapse.** Early-stop decoding measures when the correct answer first becomes readable, whereas intervention sensitivity and RQ3 readouts reveal whether later computation remains behaviorally and representationally relevant. Together, they separate *availability* from *stability*: an answer may be decodable before the latent trajectory consolidates the decision or eliminates competing modes. Through RQ2, this suggests that superposition can reflect *shared prefix computation*, where competing solutions reuse intermediate processing before divergence becomes externally visible. Probe-based and teacher-forced readouts further show that commitment is not a single event: output distributions can collapse before intermediate representations cease to carry alternatives. This aligns with RQ1’s late-step stabilization effects and with evidence that behavioral or preference directions can be isolated and steered in representation space via lightweight inference-time interventions (Rimsky et al., 2024; Li et al., 2023; Zou et al., 2023; Zhang et al., 2026). Overall, RQ1–RQ3 suggest a division of labor across latent steps, where shared intermediate computation and paradigm-dependent stabilization mechanisms support robust final-mode dominance.

**Implications for latent-reasoning design.** Although our contribution is diagnostic rather than algorithmic, the observed step-level structure suggests concrete design direc-

tions. Latent budgets should not be treated as homogeneous extra depth: computation, supervision, and regularization should be allocated selectively according to the functional role of each step. Since latent states lack natural supervision targets, future objectives could move beyond uniform CoT imitation and instead impose functional constraints on high-leverage steps or high-influence routes, such as readout consistency, perturbation robustness, downstream-state stability, or teacher-derived targets at causally important positions. The gap between early answer preference and later representational commitment further argues against forcing every step to predict the final answer: early states should preserve uncertainty or multiple candidate modes, while later states consolidate them into a perturbation-stable decision. Accordingly, stopping rules should consider not only when an answer first becomes decodable, but also when the supporting latent representation becomes stable.

## 8. Conclusion

We framed latent chain-of-thought as a step-indexed causal system and evaluated it across COCONUT and CODI with interventions, influence-structure estimation, and trajectory-level readouts. Beyond characterizing these models, our results offer design-relevant insights: latent-step budgets should be treated as an allocatable interface rather than homogeneous “extra depth,” since causal leverage concentrates unevenly and propagates along a few dominant long-range routes; and training/decoding should account for a gap between early output bias and late representational commitment, where alternatives remain latent-available even after the output distribution tilts. Together, this suggests that future latent-reasoning models can be improved by explicitly shaping where information is written and how it is consolidated across steps (e.g., encouraging more stable bottlenecks or controllable routing), rather than only scaling the number of latent steps.

## Limitations

Our conclusions are tied to specific methodological choices, including the step-level causal interface, hidden-state overwrite interventions, and teacher-forced readout. Furthermore, this study is limited to single-step edits, a fixed latent budget ( $T=6$ ), and a controlled set of paradigms, backbones, and CoT-supervised benchmarks. Although we provide supplementary checks on SIM-CoT and additional COCONUT backbones in Appendix D and Appendix E, evaluation across more paradigms, longer horizons, adaptive latent budgets, and varied intervention types is still needed to assess the generalizability of the observed patterns.

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## Impact Statement

This work studies how latent-reasoning language models can be analyzed using intervention-based measurements. The primary impact is methodological: providing tools to better diagnose, compare, and potentially improve the reliability and controllability of latent computation. While improved latent reasoning could enable more capable automated decision-making systems, the same advances may also lower the cost of generating persuasive or overconfident outputs. We therefore emphasize responsible use: evaluations should be paired with standard safety testing, and interpretations of causal measurements should avoid overclaiming model understanding beyond what the protocol supports.

## References

- Bai, X., Wu, J., Chen, Y., Wang, Z., Chen, K., Zhang, M., and Zhang, Y. Constituency Parsing using LLMs. *IEEE Transactions on Audio, Speech and Language Processing*, 33:3762–3775, 2025. doi: 10.1109/TASLPRO.2025.3600867.
- Chen, X., Zhao, A., Xia, H., Lu, X., Wang, H., Chen, Y., Zhang, W., Wang, J., Li, W., and Shen, X. Reasoning Beyond Language: A Comprehensive Survey on Latent Chain-of-Thought Reasoning. *arXiv preprint arXiv:2505.16782*, 2025. doi: 10.48550/arXiv.2505.16782. URL <https://doi.org/10.48550/arXiv.2505.16782>.
- Chrysostomou, G. and Aletras, N. Improving the faithfulness of attention-based explanations with task-specific information for text classification. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pp. 477–488, Online, August 2021. Association for Computational Linguistics. doi: 10.18653/v1/2021.acl-long.40. URL <https://aclanthology.org/2021.acl-long.40/>.
- Cobbe, K., Kosaraju, V., Bavarian, M., Chen, M., Jun, H., Kaiser, L., Plappert, M., Tworek, J., Hilton, J., Nakano, R., Hesse, C., and Schulman, J. Training Verifiers to Solve Math Word Problems. *arXiv preprint arXiv:2110.14168*, 2021. doi: 10.48550/arXiv.2110.14168. URL <https://doi.org/10.48550/arXiv.2110.14168>.
- Conmy, A., Mavor-Parker, A. N., Lynch, A., Heimersheim, S., and Garriga-Alonso, A. Towards Automated Circuit Discovery for Mechanistic Interpretability. In *Advances in Neural Information Processing Systems*, volume 36, pp. 16318–16352, New Orleans, LA, USA, December 2023. Curran Associates, Inc. URL [https://proceedings.neurips.cc/paper\\_files/paper/2023/file/34e1dbe95d34d7ebaf99b9bcaeb5b2be-Paper-Conference.pdf](https://proceedings.neurips.cc/paper_files/paper/2023/file/34e1dbe95d34d7ebaf99b9bcaeb5b2be-Paper-Conference.pdf).
- Deng, Y., Prasad, K., Fernandez, R., Smolensky, P., Chaudhary, V., and Shieber, S. Implicit Chain of Thought Reasoning via Knowledge Distillation. *arXiv preprint arXiv:2311.01460*, 2023. doi: 10.48550/arXiv.2311.01460. URL <https://doi.org/10.48550/arXiv.2311.01460>.
- Elhage, N., Hume, T., Olsson, C., Schiefer, N., Henighan, T., Kravec, S., Hatfield-Dodds, Z., Lasenby, R., Drain, D., Chen, C., Grosse, R., McCandlish, S., Kaplan, J., Amodei, D., Wattenberg, M., and Olah, C. Toy Models of Superposition. *arXiv preprint arXiv:2209.10652*, 2022. doi: 10.48550/arXiv.2209.10652. URL <https://doi.org/10.48550/arXiv.2209.10652>.
- Feder, A., Keith, K. A., Manzoor, E., Pryzant, R., Sridhar, D., Wood-Doughty, Z., Eisenstein, J., Grimmer, J., Reichart, R., Roberts, M. E., Stewart, B. M., Veitch, V., and Yang, D. Causal Inference in Natural Language Processing: Estimation, Prediction, Interpretation and Beyond. *Transactions of the Association for Computational Linguistics*, 10:1138–1158, 2022. doi: 10.1162/tacl-a-00511. URL <https://aclanthology.org/2022.tacl-1.66/>.
- Finlayson, M., Mueller, A., Gehrmann, S., Shieber, S., Linzen, T., and Belinkov, Y. Causal analysis of syntactic agreement mechanisms in neural language models. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pp. 1828–1843, Online, August 2021. Association for Computational Linguistics. doi: 10.18653/v1/2021.acl-

- long.144. URL <https://aclanthology.org/2021.acl-long.144/>.
- Geiping, J., McLeish, S. M., Jain, N., Kirchenbauer, J., Singh, S., Bartoldson, B. R., Kailkhura, B., Bhatele, A., and Goldstein, T. Scaling up Test-Time Compute with Latent Reasoning: A Recurrent Depth Approach. In *Advances in Neural Information Processing Systems*, volume 38, pp. 41340–41391, San Diego, CA, USA and Mexico City, Mexico, December 2025. Curran Associates, Inc. URL [https://proceedings.neurips.cc/paper\\_files/paper/2025/file/3b01972cf31e6fa0fe29e4b8b5c2a0a1-Paper-Conference.pdf](https://proceedings.neurips.cc/paper_files/paper/2025/file/3b01972cf31e6fa0fe29e4b8b5c2a0a1-Paper-Conference.pdf).
- Geva, M., Khashabi, D., Segal, E., Khot, T., Roth, D., and Berant, J. Did Aristotle use a laptop? a question answering benchmark with implicit reasoning strategies. *Transactions of the Association for Computational Linguistics*, 9:346–361, 2021. doi: 10.1162/tacl-a.00370. URL <https://aclanthology.org/2021.tacl-1.21/>.
- Gozeten, H. A., Ildiz, M. E., Zhang, X., Harutyunyan, H., Rawat, A. S., and Oymak, S. Continuous Chain of Thought Enables Parallel Exploration and Reasoning. In *The Fourteenth International Conference on Learning Representations*, Rio de Janeiro, Brazil, April 2026. URL <https://openreview.net/forum?id=sTPKDKn5ig>.
- Grattafiori, A., Dubey, A., Jauhri, A., et al. The Llama 3 Herd of Models. *arXiv preprint arXiv:2407.21783*, 2024. doi: 10.48550/arXiv.2407.21783. URL <https://doi.org/10.48550/arXiv.2407.21783>.
- Hao, S., Sukhbaatar, S., Su, D., Li, X., Hu, Z., Weston, J. E., and Tian, Y. Training Large Language Models to Reason in a Continuous Latent Space. In *Second Conference on Language Modeling*, Montreal, Canada, October 2025. URL <https://openreview.net/forum?id=Itxz7S4Ip3>.
- He, Y., Zheng, W., Zhu, Y., Zheng, Z., Su, L., Vasudevan, S., Guo, Q., Hong, L., and Li, J. SemCoT: Accelerating Chain-of-Thought Reasoning through Semantically-Aligned Implicit Tokens. In *Advances in Neural Information Processing Systems*, volume 38, pp. 43455–43485, San Diego, CA, USA and Mexico City, Mexico, December 2025. Curran Associates, Inc. URL [https://proceedings.neurips.cc/paper\\_files/paper/2025/file/3ddb473456a57e3cafb1ee51ddf8ff6-Paper-Conference.pdf](https://proceedings.neurips.cc/paper_files/paper/2025/file/3ddb473456a57e3cafb1ee51ddf8ff6-Paper-Conference.pdf).
- Imai, K., Keele, L., and Tingley, D. A general approach to causal mediation analysis. *Psychological Methods*, 15(4): 309–334, 2010. doi: 10.1037/a0020761.
- Kaddour, J., Lynch, A., Liu, Q., Kusner, M. J., and Silva, R. Causal Machine Learning: A Survey and Open Problems. *Foundations and Trends in Optimization*, 9(1–2):1–247, 2025. doi: 10.1561/24000000052. URL <https://doi.org/10.1561/24000000052>.
- Li, K., Patel, O., Viégas, F., Pfister, H., and Wattenberg, M. Inference-Time Intervention: Eliciting Truthful Answers from a Language Model. In *Advances in Neural Information Processing Systems*, volume 36, pp. 41451–41530, New Orleans, LA, USA, December 2023. Curran Associates, Inc. URL [https://proceedings.neurips.cc/paper\\_files/paper/2023/file/81b8390039b7302c909cb769f8b6cd93-Paper-Conference.pdf](https://proceedings.neurips.cc/paper_files/paper/2023/file/81b8390039b7302c909cb769f8b6cd93-Paper-Conference.pdf).
- McCarthy, R., SKAF, J., Ibanez-Lissen, L., Georgiev, V., Watts, C., Whittingham, H., Gonzalez-Manzano, L., Tice, C., Young, E. J., Radmard, P., and Lindner, D. Large Language Models can Learn and Generalize Steganographic Chain-of-Thought under Process Supervision. In *Advances in Neural Information Processing Systems*, volume 38, pp. 27597–27632, San Diego, CA, USA and Mexico City, Mexico, December 2025. Curran Associates, Inc. URL [https://proceedings.neurips.cc/paper\\_files/paper/2025/file/28131b22fafebba500eb7bb02e3d5b59-Paper-Conference.pdf](https://proceedings.neurips.cc/paper_files/paper/2025/file/28131b22fafebba500eb7bb02e3d5b59-Paper-Conference.pdf).
- Meng, K., Bau, D., Andonian, A., and Belinkov, Y. Locating and Editing Factual Associations in GPT. In *Advances in Neural Information Processing Systems*, volume 35, pp. 17359–17372, New Orleans, LA, USA, December 2022. Curran Associates, Inc. URL [https://proceedings.neurips.cc/paper\\_files/paper/2022/file/6f1d43d5a82a37e89b0665b33bf3a182-Paper-Conference.pdf](https://proceedings.neurips.cc/paper_files/paper/2022/file/6f1d43d5a82a37e89b0665b33bf3a182-Paper-Conference.pdf).
- Mueller, A., Brinkmann, J., Li, M., Marks, S., Pal, K., Prakash, N., Rager, C., Sankaranarayanan, A., Sharma, A. S., Sun, J., Todd, E., Bau, D., and Belinkov, Y. The Quest for the Right Mediator: Surveying Mechanistic Interpretability for NLP through the Lens of Causal Mediation Analysis. *Computational Linguistics*, 52 (1):331–378, 2026. doi: 10.1162/COLI.a.572. URL <https://doi.org/10.1162/COLI.a.572>.
- Nanda, N., Chan, L., Lieberum, T., Smith, J., and Steinhardt, J. Progress Measures for Grokking via Mechanistic

- Interpretability. In *The Eleventh International Conference on Learning Representations*, Kigali, Rwanda, May 2023. URL <https://openreview.net/forum?id=9XFSbDPmdW>.
- Pearl, J. *Causality: models, reasoning, and inference*. Cambridge University Press, Cambridge New York, NY Port Melbourne New Delhi Singapore, second edition, reprinted with corrections edition, 2000. ISBN 978-0-521-89560-6.
- Pruthi, D., Gupta, M., Dhingra, B., Neubig, G., and Lipton, Z. C. Learning to deceive with attention-based explanations. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pp. 4782–4793, Online, July 2020. Association for Computational Linguistics. doi: 10.18653/v1/2020.acl-main.432. URL <https://aclanthology.org/2020.acl-main.432/>.
- Radford, A., Wu, J., Child, R., Luan, D., Amodei, D., and Sutskever, I. Language Models are Unsupervised Multitask Learners. Technical report, OpenAI, 2019. URL [https://cdn.openai.com/better-language-models/language\\_models\\_are\\_unsupervised\\_multitask\\_learners.pdf](https://cdn.openai.com/better-language-models/language_models_are_unsupervised_multitask_learners.pdf).
- Rimsky, N., Gabrieli, N., Schulz, J., Tong, M., Hubinger, E., and Turner, A. Steering Llama 2 via Contrastive Activation Addition. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 15504–15522, Bangkok, Thailand, August 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.acl-long.828. URL <https://aclanthology.org/2024.acl-long.828/>.
- Robins, J. M. and Greenland, S. Identifiability and Exchangeability for Direct and Indirect Effects. *Epidemiology*, 3(2):143–155, 1992. URL [https://journals.lww.com/epidem/abstract/1992/03000/identifiability\\_and\\_exchangeability\\_for\\_direct\\_and.13.aspx](https://journals.lww.com/epidem/abstract/1992/03000/identifiability_and_exchangeability_for_direct_and.13.aspx).
- Ruan, Z., Li, Y., Zhu, H., Chen, Y., Li, P., Liu, Y., and Chen, G. Enhancing Large Language Model Reasoning via Selective Critical Token Fine-Tuning. *arXiv preprint arXiv:2510.10974*, 2025. doi: 10.48550/arXiv.2510.10974. URL <https://doi.org/10.48550/arXiv.2510.10974>.
- Schölkopf, B., Locatello, F., Bauer, S., Ke, N. R., Kalchbrenner, N., Goyal, A., and Bengio, Y. Toward Causal Representation Learning. *Proceedings of the IEEE*, 109(5):612–634, 2021. doi: 10.1109/JPROC.2021.3058954. URL <https://doi.org/10.1109/JPROC.2021.3058954>.
- Serrano, S. and Smith, N. A. Is attention interpretable? In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pp. 2931–2951, Florence, Italy, July 2019. Association for Computational Linguistics. doi: 10.18653/v1/P19-1282. URL <https://aclanthology.org/P19-1282/>.
- Shen, Z., Yan, H., Zhang, L., Hu, Z., Du, Y., and He, Y. CODI: Compressing Chain-of-Thought into Continuous Space via Self-Distillation. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pp. 677–693, Suzhou, China, November 2025. Association for Computational Linguistics. doi: 10.18653/v1/2025.emnlp-main.36. URL <https://aclanthology.org/2025.emnlp-main.36/>.
- Singh, R., Xu, L., and Gretton, A. Kernel Methods for Causal Functions: Dose, Heterogeneous and Incremental Response Curves. *Biometrika*, 111(2):497–516, 2024. doi: 10.1093/biomet/asad042. URL <https://doi.org/10.1093/biomet/asad042>.
- Stolfo, A., Belinkov, Y., and Sachan, M. A Mechanistic Interpretation of Arithmetic Reasoning in Language Models using Causal Mediation Analysis. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pp. 7035–7052, Singapore, December 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.emnlp-main.435. URL <https://doi.org/10.18653/v1/2023.emnlp-main.435>.
- Su, D., Zhu, H., Xu, Y., Jiao, J., Tian, Y., and Zheng, Q. Token Assorted: Mixing Latent and Text Tokens for Improved Language Model Reasoning. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pp. 57144–57163, Vancouver, BC, Canada, July 2025. PMLR. URL <https://proceedings.mlr.press/v267/su25g.html>.
- Talmor, A., Herzig, J., Lourie, N., and Berant, J. CommonsenseQA: A question answering challenge targeting commonsense knowledge. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pp. 4149–4158, Minneapolis, Minnesota, June 2019. Association for Computational Linguistics. doi: 10.18653/v1/N19-1421. URL <https://aclanthology.org/N19-1421/>.

- Thulasiraman, K. and Swamy, M. N. S. *Graphs: theory and algorithms*. John Wiley & Sons, Inc., USA, 1992. ISBN 0471513563.
- Turpin, M., Michael, J., Perez, E., and Bowman, S. R. Language Models don't Always Say What They Think: Unfaithful Explanations in Chain-of-Thought Prompting. In *Advances in Neural Information Processing Systems*, volume 36, pp. 74952–74965, New Orleans, LA, USA, December 2023. Curran Associates, Inc. URL [https://proceedings.neurips.cc/paper\\_files/paper/2023/file/ed3fea9033a80fea1376299fa7863f4a-Paper-Conference.pdf](https://proceedings.neurips.cc/paper_files/paper/2023/file/ed3fea9033a80fea1376299fa7863f4a-Paper-Conference.pdf).
- Tutek, M., Hashemi Chaleshtori, F., Marasovic, A., and Belinkov, Y. Measuring Chain of Thought Faithfulness by Unlearning Reasoning Steps. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pp. 9935–9960, Suzhou, China, November 2025. Association for Computational Linguistics. doi: 10.18653/v1/2025.emnlp-main.504. URL <https://aclanthology.org/2025.emnlp-main.504/>.
- Vig, J., Gehrmann, S., Belinkov, Y., Qian, S., Nevo, D., Singer, Y., and Shieber, S. Investigating Gender Bias in Language Models Using Causal Mediation Analysis. In *Advances in Neural Information Processing Systems*, volume 33, pp. 12388–12401, virtual, December 2020. Curran Associates, Inc. URL [https://proceedings.neurips.cc/paper\\_files/paper/2020/file/92650b2e92217715fe312e6fa7b90d82-Paper.pdf](https://proceedings.neurips.cc/paper_files/paper/2020/file/92650b2e92217715fe312e6fa7b90d82-Paper.pdf).
- Wang, F., Mo, W., Wang, Y., Zhou, W., and Chen, M. A causal view of entity bias in (large) language models. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pp. 15173–15184, Singapore, December 2023a. Association for Computational Linguistics. doi: 10.18653/v1/2023.findings-emnlp.1013. URL <https://aclanthology.org/2023.findings-emnlp.1013/>.
- Wang, K. R., Variengien, A., Conmy, A., Shlegeris, B., and Steinhardt, J. Interpretability in the Wild: A Circuit for Indirect Object Identification in GPT-2 Small. In *The Eleventh International Conference on Learning Representations*, Kigali, Rwanda, May 2023b. URL <https://openreview.net/forum?id=NpsVSN6o4ul>.
- Wang, W., Wang, Y., Chen, K., and Huang, H. Beyond Majority Voting: Towards Fine-grained and More Reliable Reward Signal for Test-Time Reinforcement Learning. *arXiv preprint arXiv:2512.15146*, 2025. doi: 10.48550/arXiv.2512.15146. URL <https://doi.org/10.48550/arXiv.2512.15146>.
- Wei, J., Wang, X., Schuurmans, D., Bosma, M., Ichter, B., Xia, F., Chi, E. H., Le, Q. V., and Zhou, D. Chain-of-Thought Prompting Elicits Reasoning in Large Language Models. In *Advances in Neural Information Processing Systems*, volume 35, pp. 24824–24837, New Orleans, LA, USA, December 2022. Curran Associates, Inc. URL [https://proceedings.neurips.cc/paper\\_files/paper/2022/file/9d5609613524ecf4f15af0f7b31abca4-Paper-Conference.pdf](https://proceedings.neurips.cc/paper_files/paper/2022/file/9d5609613524ecf4f15af0f7b31abca4-Paper-Conference.pdf).
- Wei, X., Liu, X., Zang, Y., Dong, X., Cao, Y., Wang, J., Qiu, X., and Lin, D. SIM-CoT: Supervised Implicit Chain-of-Thought. In *The Fourteenth International Conference on Learning Representations*, Rio de Janeiro, Brazil, April 2026. URL <https://openreview.net/forum?id=6YRJ4jmVQ1>.
- Wu, H., Teng, Z., and Tu, K. Parallel Continuous Chain-of-Thought with Jacobi Iteration. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pp. 914–926, Suzhou, China, November 2025. Association for Computational Linguistics. doi: 10.18653/v1/2025.emnlp-main.47. URL <https://aclanthology.org/2025.emnlp-main.47/>.
- Wu, J., Lu, J., Ren, Z., Hu, G., Wu, Z., Dai, D., and Wu, H. LLMs are Single-threaded Reasoners: Demystifying the Working Mechanism of Soft Thinking. In *The Fourteenth International Conference on Learning Representations*, Rio de Janeiro, Brazil, April 2026. URL <https://openreview.net/forum?id=ASLuOoP78o>.
- Xiao, Z., Dou, D., Xiong, B., Chen, Y., and Chen, G. Enhancing Uncertainty Estimation in LLMs with Expectation of Aggregated Internal Belief. *arXiv preprint arXiv:2509.01564*, 2025. doi: 10.48550/arXiv.2509.01564. URL <https://doi.org/10.48550/arXiv.2509.01564>.
- Xu, Y., Guo, X., Zeng, Z., and Miao, C. SoftCoT: Soft Chain-of-Thought for Efficient Reasoning with LLMs. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 23336–23351, Vienna, Austria, July 2025. Association for Computational Linguistics. doi: 10.18653/v1/2025.acl-long.1137. URL <https://aclanthology.org/2025.acl-long.1137/>.
- Yan, Y., Jiang, L., Jiang, J., Li, S., Wen, Z., Zhang, Z., Zhou, J., Shao, J., Zhuang, Y., and Shen, Y. InftyThink+: Effective and Efficient Infinite-Horizon Reasoning via Reinforcement Learning. *arXiv preprint arXiv:2602.06960*,

2026. doi: 10.48550/arXiv.2602.06960. URL <https://doi.org/10.48550/arXiv.2602.06960>.
- Yang, A., Li, A., Yang, B., Zhang, B., Hui, B., Zheng, B., Yu, B., Gao, C., Huang, C., Lv, C., Zheng, C., Liu, D., Zhou, F., Huang, F., Hu, F., Ge, H., Wei, H., Lin, H., Tang, J., Yang, J., Tu, J., Zhang, J., Yang, J., Yang, J., Zhou, J., Zhou, J., Lin, J., Dang, K., Bao, K., Yang, K., Yu, L., Deng, L., Li, M., Xue, M., Li, M., Zhang, P., Wang, P., Zhu, Q., Men, R., Gao, R., Liu, S., Luo, S., Li, T., Tang, T., Yin, W., Ren, X., Wang, X., Zhang, X., Ren, X., Fan, Y., Su, Y., Zhang, Y., Zhang, Y., Wan, Y., Liu, Y., Wang, Z., Cui, Z., Zhang, Z., Zhou, Z., and Qiu, Z. Qwen3 Technical Report. *arXiv preprint arXiv:2505.09388*, 2025. doi: 10.48550/arXiv.2505.09388. URL <https://doi.org/10.48550/arXiv.2505.09388>.
- Yang, Z., Liu, Y., and Ouyang, C. Causal intervention-based few-shot named entity recognition. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pp. 15635–15646, Singapore, December 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.findings-emnlp.1046. URL <https://aclanthology.org/2023.findings-emnlp.1046/>.
- Yao, L., Chu, Z., Li, S., Li, Y., Gao, J., and Zhang, A. A survey on causal inference. *ACM Transactions on Knowledge Discovery from Data*, 15(5):1–46, 2021. doi: 10.1145/3444944. URL <https://doi.org/10.1145/3444944>.
- Yu, X., Wang, Z., Yang, L., Li, H., Liu, A., Xue, X., Wang, J., and Yang, M. Causal Sufficiency and Necessity Improves Chain-of-Thought Reasoning. In *Advances in Neural Information Processing Systems*, volume 38, pp. 126109–126141, San Diego, CA, USA and Mexico City, Mexico, December 2025. Curran Associates, Inc. URL [https://papers.nips.cc/paper\\_files/paper/2025/file/b7870bd43b2d133aed95582ae5d82a4-Paper-Conference.pdf](https://papers.nips.cc/paper_files/paper/2025/file/b7870bd43b2d133aed95582ae5d82a4-Paper-Conference.pdf).
- Zhang, K. et al. A Survey of Reinforcement Learning for Large Reasoning Models. *arXiv preprint arXiv:2509.08827*, 2025a. doi: 10.48550/arXiv.2509.08827. URL <https://doi.org/10.48550/arXiv.2509.08827>.
- Zhang, Y., Ma, J., Hou, Y., Bai, X., Chen, K., Xiang, Y., Yu, J., and Zhang, M. Evaluating and Steering Modality Preferences in Multimodal Large Language Model. In *Proceedings of the 43rd International Conference on Machine Learning*, Seoul, South Korea, July 2026. doi: 10.48550/arXiv.2505.20977. URL <https://doi.org/10.48550/arXiv.2505.20977>.
- Zhang, Z., He, X., Yan, W., Shen, A., Zhao, C., and Wang, X. Soft Thinking: Unlocking the Reasoning Potential of LLMs in Continuous Concept Space. In *Advances in Neural Information Processing Systems*, volume 38, pp. 168990–169012, San Diego, CA, USA and Mexico City, Mexico, December 2025b. Curran Associates, Inc. URL [https://proceedings.neurips.cc/paper\\_files/paper/2025/file/f7396d1c54d51416958d63e285377103-Paper-Conference.pdf](https://proceedings.neurips.cc/paper_files/paper/2025/file/f7396d1c54d51416958d63e285377103-Paper-Conference.pdf).
- Zhu, H., Hao, S., Hu, Z., Jiao, J., Russell, S. J., and Tian, Y. Reasoning by Superposition: A Theoretical Perspective on Chain of Continuous Thought. In *Advances in Neural Information Processing Systems*, volume 38, pp. 79931–79963, San Diego, CA, USA and Mexico City, Mexico, December 2025a. Curran Associates, Inc. URL [https://proceedings.neurips.cc/paper\\_files/paper/2025/file/72c363c2a573ca2128bd176d3317696b-Paper-Conference.pdf](https://proceedings.neurips.cc/paper_files/paper/2025/file/72c363c2a573ca2128bd176d3317696b-Paper-Conference.pdf).
- Zhu, R.-J., Peng, T., Cheng, T., Qu, X., Huang, J., Zhu, D., Wang, H., Xue, K., Zhang, X., Shan, Y., Cai, T., Kergan, T., Kembay, A., Smith, A., Lin, C., Nguyen, B., Pan, Y., Chou, Y., Cai, Z., Wu, Z., Zhao, Y., Liu, T., Yang, J., Zhou, W., Zheng, C., Li, C., Zhou, Y., Li, Z., Zhang, Z., Liu, J., Zhang, G., Huang, W., and Eshraghian, J. A Survey on Latent Reasoning. *arXiv preprint arXiv:2507.06203*, 2025b. doi: 10.48550/arXiv.2507.06203. URL <https://doi.org/10.48550/arXiv.2507.06203>.
- Zhu, R.-J., Wang, Z., Hua, K., Zhang, T., Li, Z., Que, H., Wei, B., Wen, Z., Yin, F., Xing, H., Li, L., Shi, J., Ma, K., Li, S., Kergan, T., Smith, A., Qu, X., Hui, M., Wu, B., Min, Q., Huang, H., Zhou, X., Ye, W., Liu, J., Yang, J., Shi, Y., Lin, C., Zhao, E., Cai, T., Zhang, G., Huang, W., Bengio, Y., and Eshraghian, J. Scaling Latent Reasoning via Looped Language Models. *arXiv preprint arXiv:2510.25741*, 2025c. doi: 10.48550/arXiv.2510.25741. URL <https://doi.org/10.48550/arXiv.2510.25741>.
- Zou, A., Phan, L., Chen, S., Campbell, J., Guo, P., Ren, R., Pan, A., Yin, X., Mazeika, M., Dombrowski, A.-K., Goel, S., Li, N., Byun, M. J., Wang, Z., Mallen, A., Basart, S., Koyejo, S., Song, D., Fredrikson, M., Kolter, J. Z., and Hendrycks, D. Representation Engineering: A Top-Down Approach to AI Transparency. *arXiv preprint arXiv:2310.01405*, 2023. doi: 10.48550/arXiv.2310.01405. URL <https://doi.org/10.48550/arXiv.2310.01405>.

## A. Implementation Details

### A.1. Training settings for reproduced COCONUT and CODI

We reproduce COCONUT across three backbones (GPT-2, Llama3-1B, and Qwen3-4B-Instruct) and three datasets (GSM8K, CommonsenseQA, and StrategyQA) using the standard stage-wise latent-replacement curriculum. On GSM8K, we follow the official two-stage recipe (*CoT-SFT*  $\rightarrow$  *Coconut*): the model is first trained with explicit CoT supervision, and then continued with Coconut-style latent reasoning initialized from the CoT checkpoint. For GPT-2, both stages are trained for 25 epochs (25+25). For larger backbones, we use a shorter schedule (5+10) consistent with the large-model setting. On CommonsenseQA and StrategyQA, we adopt a uniform latent curriculum that progressively increases the latent-step budget up to  $T=6$  using one epoch per latent stage (6 epochs total), followed by 4 additional epochs at the final stage (i.e., 6+4). Across backbones, we use AdamW-style optimization with weight decay 0.01 and a backbone-dependent learning rate: GPT-2 runs use  $\text{lr} = 10^{-4}$ , while Llama/Qwen runs use  $\text{lr} = 10^{-5}$ .

For CODI, we use *officially released* checkpoints for GSM8K on GPT-2 and Llama3-1B. All other CODI results in this paper are obtained from our reproduced checkpoints trained on the remaining dataset/backbone combinations. We follow the official CODI training pipeline (LoRA-based fine-tuning with a fixed number of latent tokens, cosine learning-rate scheduling with warmup, mixed-precision training, and the projection head enabled in our runs), while allowing dataset/backbone-dependent learning rates for stable optimization. Concretely, our reproduced CODI runs use learning rates in the range  $[5 \times 10^{-6}, 3 \times 10^{-3}]$  depending on the backbone and dataset (e.g.,  $3 \times 10^{-3}$  for GPT-2,  $8 \times 10^{-4}$  for Llama-1B, and  $2 \times 10^{-4}/10^{-5}/5 \times 10^{-6}$  for Qwen3-4B-Instruct on GSM8K-Aug/CommonsenseQA/StrategyQA, respectively). This yields CODI checkpoints matched to our Coconut reproductions in latent-step budget ( $T=6$ ) and task coverage, enabling controlled comparisons under identical intervention and readout protocols.

### A.2. Intervention operators: robustness and choice of zero overwrite

Our step-wise necessity analysis (RQ1) instantiates the single-step do-intervention (Def. 3.1) via a concrete *intervention operator* that maps a realized latent state  $h_t$  to an edited state  $\tilde{h}_t$ . To verify that our conclusions are not tied to a particular operator, we compare six commonly used perturbations that preserve the same causal interface: overwrite one step, then recompute downstream computation under fixed  $x$  and  $\theta$ . Because activation-level interventions can be sensitive to the choice of corruption or replacement operator, we treat operator variants as robustness checks rather than as uniquely identified causal estimands. This follows the caution used in activation patching, causal tracing, and mediator-search work, where measured effects can depend on the chosen mediator, intervention operator, and readout protocol (Meng et al., 2022; Conmy et al., 2023; Mueller et al., 2026).

Table 1. Intervention operators used in the robustness check. Here,  $\mu$  denotes the global mean latent state,  $\mu_t$  denotes the step-specific mean at step  $t$ ,  $\epsilon \sim \mathcal{N}(0, I)$ , and  $\sigma$  is a fixed noise scale.

| Operator         | Description                                  | Edited state $\tilde{h}_t$ |
|------------------|----------------------------------------------|----------------------------|
| zero             | Replace latent with zeros                    | <b>0</b>                   |
| mean             | Replace with global mean                     | $\mu$                      |
| mean.step        | Replace with step-specific mean              | $\mu_t$                    |
| gaussian_h       | Add Gaussian noise to $h_t$                  | $h_t + \sigma\epsilon$     |
| gaussian_mu      | Add Gaussian noise around global mean        | $\mu + \sigma\epsilon$     |
| gaussian_mu.step | Add Gaussian noise around step-specific mean | $\mu_t + \sigma\epsilon$   |

Figures 9a and 9b show two representative examples on CODI (Llama3-1B) for GSM8K and CommonsenseQA, respectively. Each heatmap cell reports the **flip rate**  $\text{Flip}(t)$  used throughout RQ1, computed as the fraction of examples whose final decoded prediction changes under the intervention at step  $t$ . Following our main metric, we aggregate both wrong $\rightarrow$ right and right $\rightarrow$ wrong flips (i.e., any decision change relative to the baseline rollout), so larger values indicate stronger decision-level dependence on the intervened step. Across operators, the qualitative step-wise sensitivity patterns are stable: operators that substantially perturb the latent state yield similar relative trends over steps, while weaker/noisier operators typically reduce absolute flip rates without altering the overall step profile. This robustness indicates that our RQ1 findings are not driven by a specific perturbation choice.

We adopt **zero overwrite** (`zero`) as the default operator for the main paper for two practical reasons. First, it is deterministic and parameter-free, eliminating tuning choices (e.g., the noise scale  $\sigma$ ) and reducing variance across runs. Second, it applies

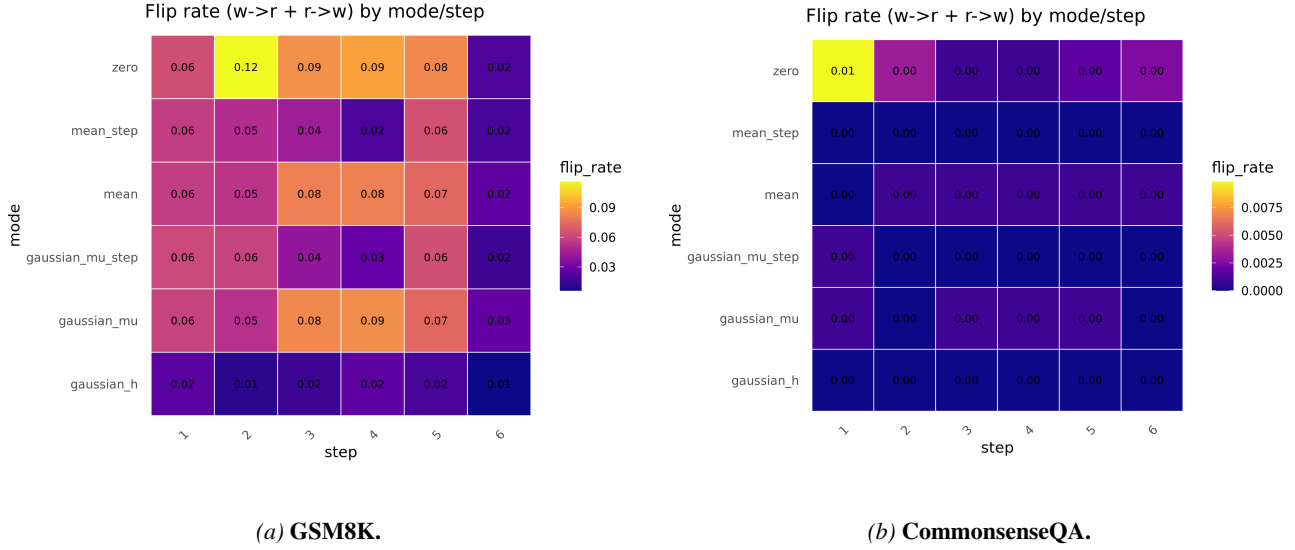


Figure 9. **Intervention-operator comparison (CODI Llama3-1B).** Each cell shows the flip rate  $\text{Flip}(t)$ , the fraction of examples whose decoded final prediction changes when intervening at step  $t$  aggregated over wrong→right and right→wrong flips.

uniformly across architectures and training recipes, making cross-model comparisons more fair: the intervention strength does not depend on backbone-specific hidden-state norms or distributional statistics beyond the shared representation space. We therefore use `zero` throughout the main experiments for reproducibility and interpretability, and treat the remaining operators as sanity checks that validate the stability of our conclusions.

### A.3. Teacher-forced readouts and influence matrix

To measure propagation effects with minimal sampling noise, we use teacher-forced readouts on a canonical gold-answer string. The answer template follows each method’s training paradigm (rather than the dataset): for COCONUT we use “[prefix] ### {answer}”, while for CODI we use “[prefix] The answer is {answer}”. We denote the resulting gold answer token sequence by  $a_{1:L}$ .

**Teacher-forced distributions.** Given a trajectory (baseline or intervened) and a designated readout step  $s$ , we compute teacher-forced logits for the gold answer tokens  $a_{1:L}$  and obtain token-level predictive distributions

$$p^{(s)}(a_\ell \mid a_{<\ell}, x) = \text{softmax}(\mathbf{z}_\ell^{(s)}), \quad \ell = 1, \dots, L, \quad (12)$$

where  $\mathbf{z}_\ell^{(s)}$  denotes the teacher-forced logit vector at gold token position  $\ell$  when reading out from step  $s$  (with the same  $x$  and fixed parameters  $\theta$ ). We compute these distributions for both the baseline rollout ( $p_{\text{base}}^{(s)}$ ) and the intervened rollout ( $p_{\text{int}}^{(s)}$ ).

**Influence matrix from token-averaged KL.** For an intervention applied at step  $t$  and a readout at step  $s$ , we define the influence weight  $W_{t,s}$  (Eq. 10) as the token-averaged KL divergence between the baseline and intervened teacher-forced distributions on the gold answer:

$$W_{t,s} = \frac{1}{L} \sum_{\ell=1}^L D_{\text{KL}} \left( p_{\text{base}}^{(s)}(\cdot \mid a_{<\ell}, x) \parallel p_{\text{int}}^{(s)}(\cdot \mid a_{<\ell}, x) \right). \quad (13)$$

Intuitively,  $W_{t,s}$  measures how much an intervention at step  $t$  changes the model’s predictive distribution over the gold answer when the trajectory is read out at step  $s$ . We aggregate  $W_{t,s}$  over examples by averaging across the evaluation set.

**Visualization and sparsification.** To visualize influence graphs, we apply the same sparsification protocol as in the main text. Specifically, we threshold edges at  $\alpha \cdot \max(W)$  with  $\alpha = 0.1$ , and for each source step we retain only the top-1

outgoing edge for readability. Edge weights in figures correspond to the (dataset-averaged) dense matrix entries  $W_{t,s}$  prior to sparsification.

**Why teacher forcing.** Teacher forcing isolates propagation effects from output sampling variability: it evaluates changes in the model’s conditional distribution along a fixed gold answer path, rather than changes in a sampled decoded string. This makes  $W$  a more stable proxy for step-to-step influence, especially when the decoded answer is short or when sampling induces high variance across rollouts.

## B. Dataset Information

### B.1. Dataset Statistics

Table 2 reports the dataset split sizes used in our experiments. All evaluations use the original benchmark test sets from the corresponding dataset papers. For training, we use CoT-augmented variants when required by the training paradigms: for GSM8K we train on GSM8K-Aug from (Deng et al., 2023) while evaluating on the original GSM8K test set (Cobbe et al., 2021); for CommonsenseQA and StrategyQA, we use CoT-augmented training data released by CODI (Shen et al., 2025) and evaluate on the original test sets (Talmor et al., 2019; Geva et al., 2021). These augmentations affect training-time supervision only; we do not modify any benchmark test set.

*Table 2. Dataset statistics used in our experiments.* Train sizes correspond to the training splits actually used for model training (GSM8K-Aug (Deng et al., 2023); CommonsenseQA-CoT and StrategyQA-CoT from CODI (Shen et al., 2025)). Test sizes correspond to the original benchmark test splits (Talmor et al., 2019; Cobbe et al., 2021; Geva et al., 2021).

| DATASET       | TRAIN   | TEST  |
|---------------|---------|-------|
| COMMONSENSEQA | 8,096   | 1,221 |
| GSM8K         | 385,620 | 1,319 |
| STRATEGYQA    | 1,809   | 229   |

### B.2. Dataset Examples

The following examples illustrate the CoT-augmented training format used in our runs. CoT rationales are *not* part of the original benchmarks; they are training-time supervision taken from the corresponding augmented variants (GSM8K-Aug (Deng et al., 2023) and CODI-released CoT data (Shen et al., 2025)). We evaluate on the original benchmark test sets without modification.

#### GSM8K-Aug (training)

**Question:** Out of 600 employees in a company, 30% got promoted while 10% received bonus. How many employees did not get either a promotion or a bonus?

**CoT:**  $600 \times 30/100 = 180$ ,  $600 \times 10/100 = 60$ ,  $600 - 240 = 360$

**Answer:** 360

#### CommonsenseQA-CoT (training)

**Question:** Google Maps and other highway and street GPS services have replaced what? Choices: A: united states B: mexico C: countryside D: atlas E: oceans

**CoT:** Google Maps and other GPS services are tools used for navigation and mapping, which have largely replaced traditional paper maps and atlases that people used in the past for similar purposes. The other options, such as countries and geographic features, do not relate to navigational tools. Thus, the best answer is D: atlas.

**Answer:** D

## StrategyQA-CoT (training)

**Question:** Would a dog respond to bell before Grey seal?

**CoT:** To determine if a dog would respond to a bell before a Grey seal, we need to consider the context and conditioning involved. Dogs are often trained to respond to bells due to their domestication and close relationships with humans. Classic conditioning experiments, like Pavlov’s dogs, indicate that dogs can be trained to associate a bell with specific actions, such as being fed. Grey seals, on the other hand, are wild marine mammals that are not typically subjected to the same kind of training or conditioning that dogs regularly undergo. Therefore, without specific conditioning, a Grey seal is unlikely to naturally associate or respond to a bell in the same way a trained dog would. Thus, a dog, particularly one that has been conditioned, is expected to respond to a bell before a Grey seal.

**Answer:** true

## C. Additional RQ2 Details and Robustness Checks

### C.1. Dense adjacency matrices for influence graphs

The main text visualizes sparsified influence graphs for readability (threshold  $\alpha=0.1$  and top-1 outgoing edge per node). Here we provide the corresponding dense influence matrices  $W$  (Eq. 10) as heatmaps, which are the objects used to compute all structure metrics.

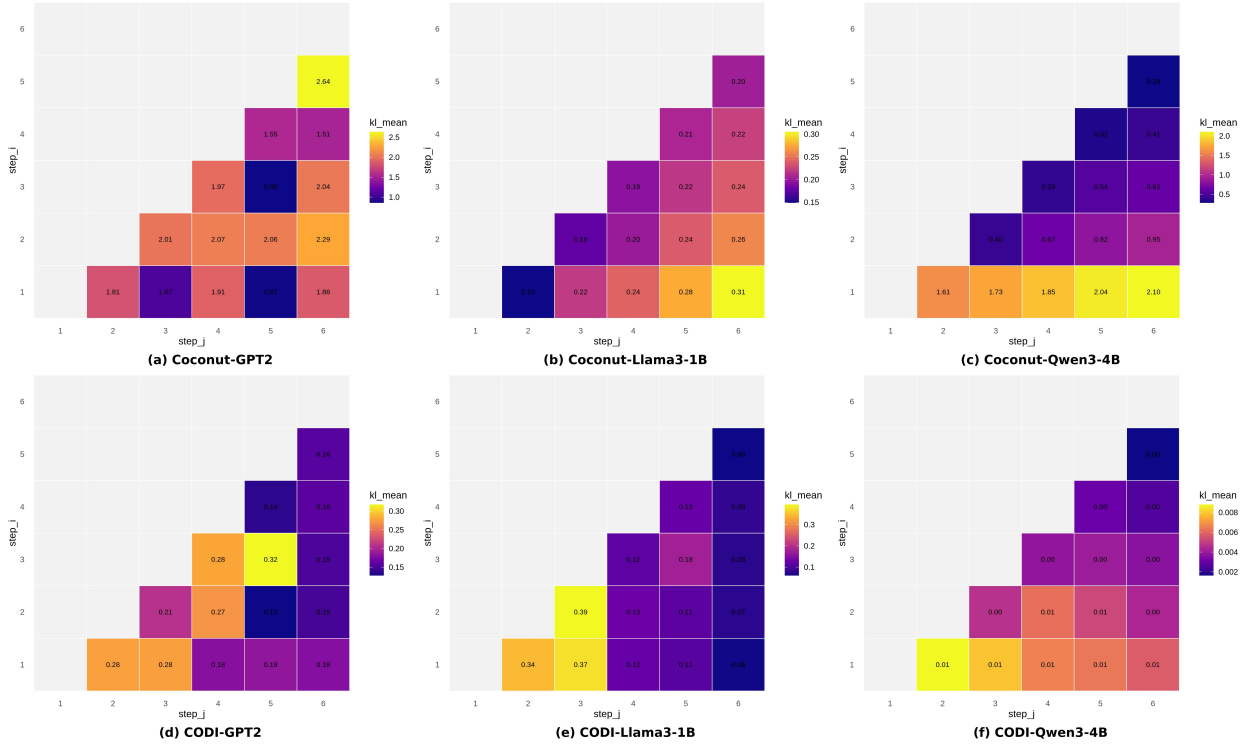


Figure 10. Dense influence matrices for latent reasoning (GSM8K). Each cell  $(t, s)$  shows  $W_{t,s}$  from Eq. 10 (teacher-forced KL shift on the gold answer when intervening at  $t$  and reading out at  $s$ ).

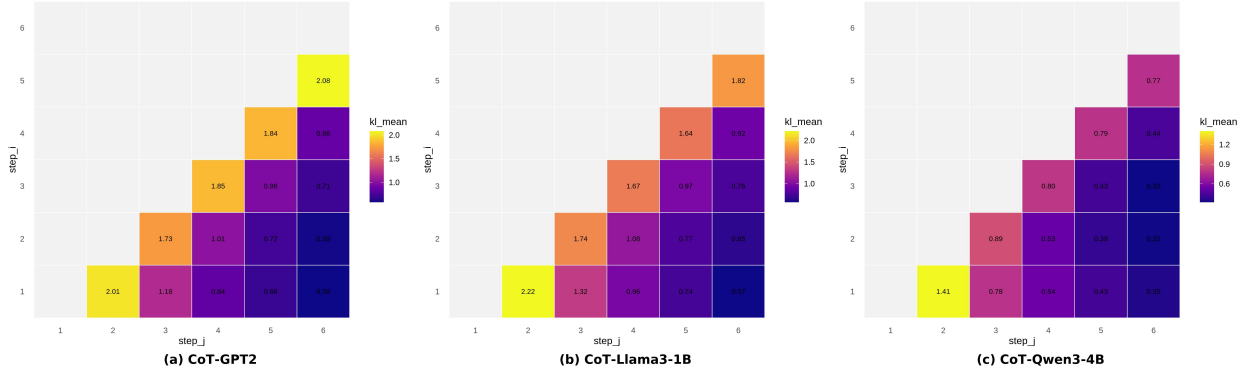


Figure 11. Dense influence matrices for explicit CoT (GSM8K; CoT-SFT). Each cell  $(t, s)$  shows  $W_{t,s}$  from Eq. 10 computed on segmented CoT-step states.

## C.2. Additional CommonsenseQA results

This subsection provides additional RQ2 results on CommonsenseQA, including dense influence matrices (latent and explicit), sparsified principal influence graphs, and the corresponding structure metrics. All quantities follow the same construction as in Sec. 5, based on the dense matrix  $W$  from Eq. 10.

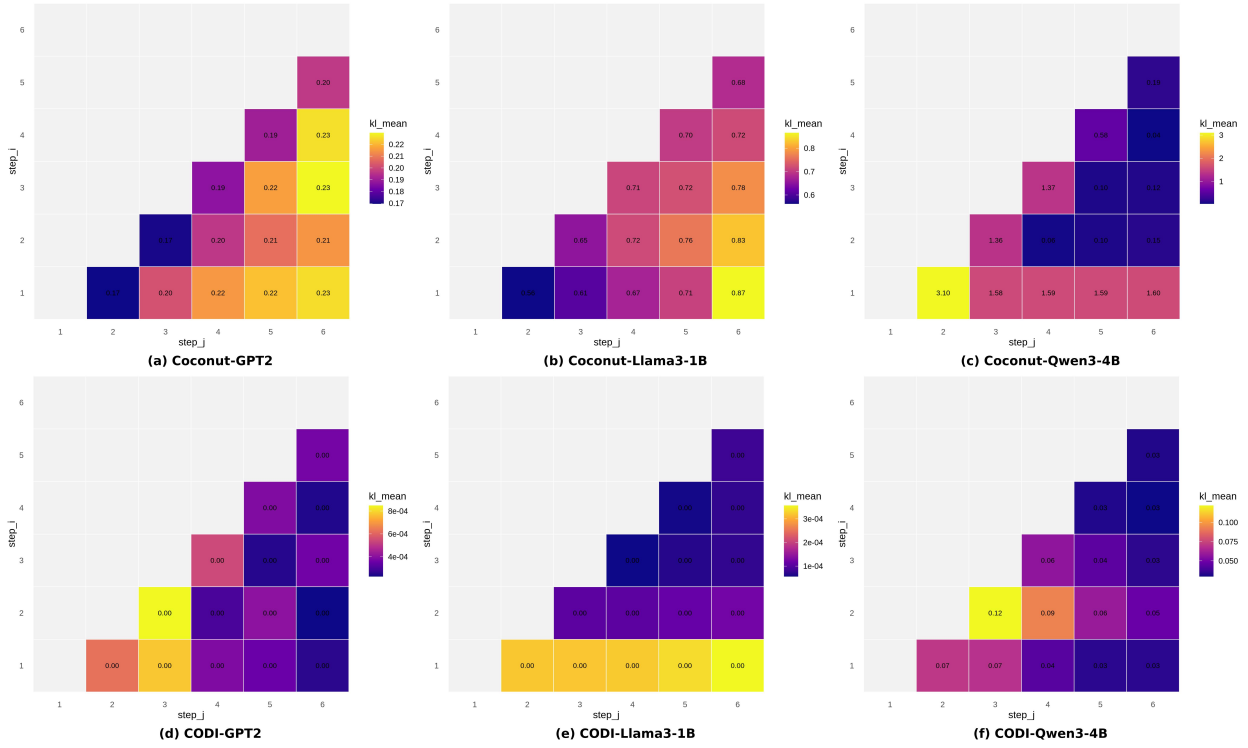


Figure 12. Dense influence matrices for latent reasoning (CommonsenseQA; COCONUT/CODI). Each cell  $(t, s)$  shows  $W_{t,s}$  from Eq. 10 under teacher-forced readouts.

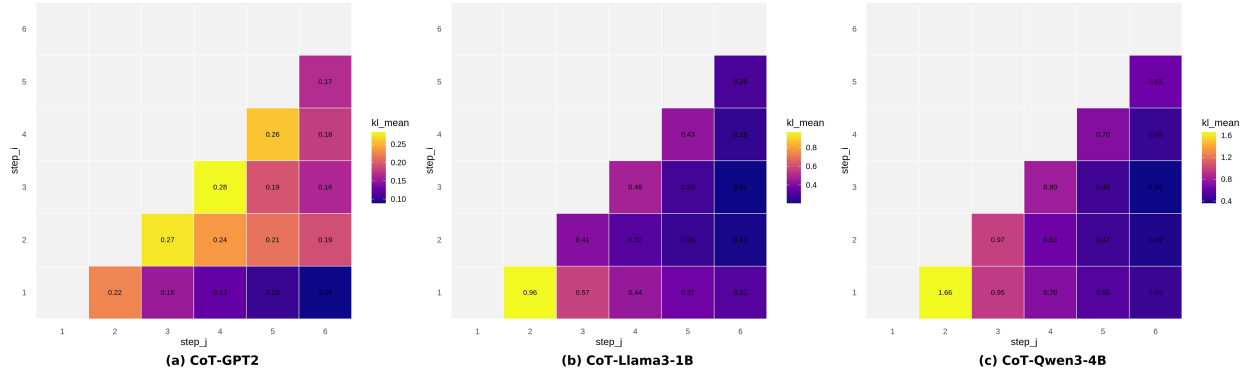


Figure 13. Dense influence matrices for explicit CoT (CommonsenseQA; CoT-SFT). Each cell  $(t, s)$  shows  $W_{t,s}$  from Eq. 10 computed on segmented CoT-step states.

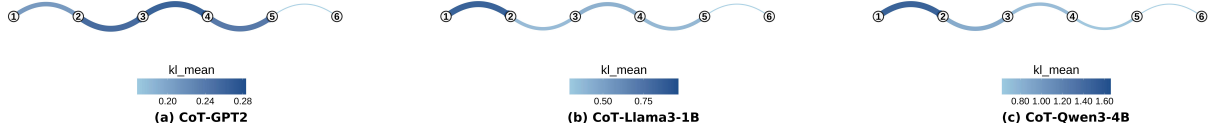


Figure 14. Explicit CoT principal influence graphs (CommonsenseQA; CoT-SFT baselines). Nodes denote the first  $T=6$  segmented CoT steps. Edge  $t \rightarrow s$  indicates propagation strength  $W_{t,s}$  from Eq. 10. For readability, we show only top-1 outgoing edges after thresholding at  $\alpha=0.1 \cdot \max(W)$ .

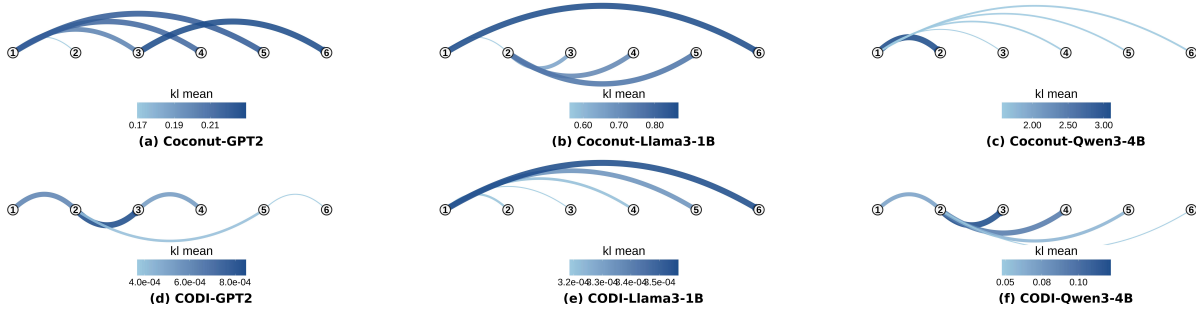


Figure 15. Latent principal influence graphs (CommonsenseQA; COCONUT/CODI). Nodes are latent steps  $t \in \{1, \dots, 6\}$ . Edge weights follow Eq. 10 under single-step interventions, rendered with the same sparsification protocol as Figure 14.

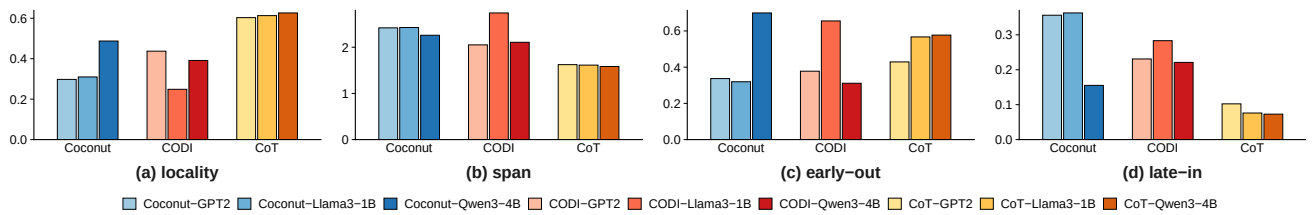


Figure 16. Structure metrics on influence graphs (CommonsenseQA). Metrics are computed on the normalized matrix  $\bar{W}$  (Eq. 14) and use the same hyperparameters as the main text.

### C.3. Definitions of structure metrics

We compute structure metrics on a normalized influence matrix  $\bar{W}$  obtained by re-scaling  $W$  over valid entries ( $t < s$ ):

$$\bar{W}_{t,s} = \frac{W_{t,s}}{\sum_{a < b} W_{a,b} + \epsilon}. \quad (14)$$

Let  $\mathcal{E} = \{(t, s) : 1 \leq t < s \leq T\}$ . We define:

$$\text{Locality}(k) = \sum_{(t,s) \in \mathcal{E}} \mathbf{1}\{s - t \leq k\} \bar{W}_{t,s}, \quad (15)$$

$$\text{Span} = \sum_{(t,s) \in \mathcal{E}} (s - t) \bar{W}_{t,s}, \quad (16)$$

$$\text{EarlyOut}(m) = \sum_{t \leq m} \sum_{s > t} \bar{W}_{t,s}, \quad (17)$$

$$\text{LateIn}(m) = \sum_{s \geq m} \sum_{t < s} \bar{W}_{t,s}. \quad (18)$$

In all experiments we use  $k=1$  and  $m=2$  (early) /  $m=5$  (late) for  $T=6$ .

### C.4. Influence structures under alternative intervention operators

Appendix A.2 verifies that the RQ1 necessity profiles are qualitatively stable across six intervention operators. We additionally test whether the RQ2 influence structure is specific to zero overwrite. We recompute the influence matrix  $W$  from Eq. 10 on GSM8K using two non-zero operators: global-mean replacement (`mean`) and Gaussian perturbation around the realized latent state (`gaussian_h`). Both operators preserve the same causal interface as the main experiment: exactly one latent step is modified, downstream computation is recomputed, and propagation is measured by teacher-forced KL at downstream readout steps.

Figure 17 shows the resulting principal influence graphs for COCONUT-GPT2 and CODI-GPT2. The absolute edge magnitudes vary with the intervention strength, but the qualitative conclusion is unchanged: influence is non-uniform across steps and is not restricted to adjacent transitions. In particular, non-local routes remain visible under both global-mean replacement and Gaussian perturbation, indicating that the skip-dominant RQ2 pattern is not an artifact of zero overwrite alone.

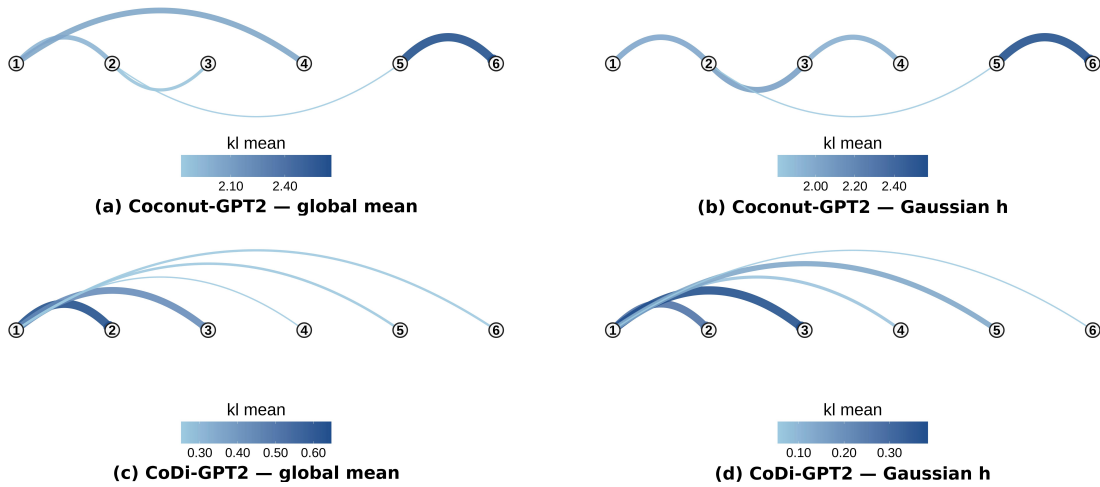


Figure 17. **RQ2 operator robustness on GSM8K.** We recompute principal influence graphs for COCONUT-GPT2 and CODI-GPT2 using global-mean replacement (`mean`) and Gaussian perturbation around the realized latent state (`gaussian_h`). Edge weights are computed with the same teacher-forced KL readout as Eq. 10. Non-local influence remains visible under both operators.

### C.5. Readout-conditioned influence structures

The main RQ2 experiment uses answer-level teacher-forced KL because it directly measures how an upstream intervention changes the model’s distribution over the gold answer path. To check whether the influence graph depends on this specific readout, we also evaluate a hidden-state LM-head readout that does not condition on the gold answer string. For an intervention at step  $t$  and a downstream readout step  $s$ , we project the baseline and intervened hidden states at step  $s$  through the language-model head and compute

$$W_{t,s}^{LM} = \mathbb{E}_t \left[ D_{KL} \left( \text{softmax}(\text{LMHead}(h_{i,s})) \parallel \text{softmax}(\text{LMHead}(\tilde{h}_{i,s}^{\text{do}(t)})) \right) \right], \quad t < s. \quad (19)$$

This readout asks whether the perturbation changes the one-step vocabulary distribution directly induced by the downstream latent state, rather than the teacher-forced distribution along a fixed answer template.

Figure 18 reports the resulting GSM8K influence graphs. The graphs remain non-uniform, and several models still exhibit long-range cross-step influence. This supports the interpretation of RQ2 as a protocol-conditioned empirical influence structure, while showing that the main non-locality finding is not tied to a single answer-template readout.

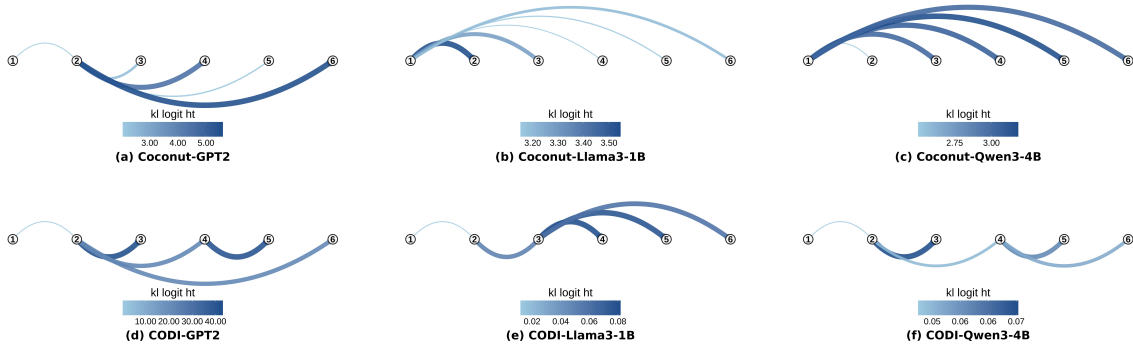


Figure 18. **RQ2 readout robustness on GSM8K.** Influence graphs computed with the hidden-state LM-head readout in Eq. 19, rather than answer-level teacher-forced KL. The main qualitative pattern remains: influence is non-uniform and often non-local across latent steps.

### D. Additional Paradigm Analysis

To test whether the observations are limited to COCONUT and CODI, we additionally evaluate SIM-CoT (Wei et al., 2026) on GSM8K using the same  $T=6$  step interface where available. This experiment is intended as a paradigm-coverage check, not as a claim that the three paradigms exhaust latent reasoning methods.

Figure 19 shows the RQ1 flip-rate profiles. The Sim-CoT variants also show heterogeneous step importance: the flip rate is not flat across latent steps, and some steps are substantially more decision-relevant than others.

Figure 20 reports the corresponding RQ2 principal influence graphs. Compared with the skip-heavy patterns observed for several COCONUT/CODI models, Sim-CoT exhibits relatively more local connectivity in these runs. This difference is consistent with its step-aligned supervision objective: stronger step-level alignment can make adjacent latent transitions more prominent, even though the overall influence structure remains non-uniform.

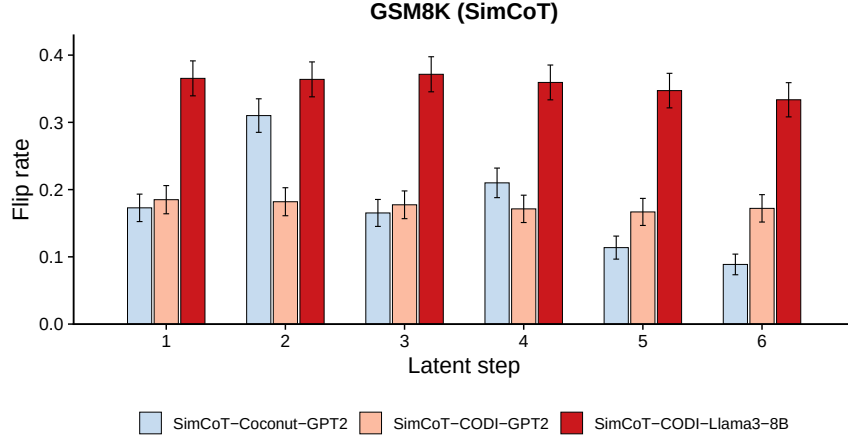


Figure 19. **RQ1 necessity check for SIM-CoT on GSM8K.** Flip rates are computed under the same single-step intervention protocol as Figure 3. Step importance remains heterogeneous under an additional latent-reasoning paradigm.

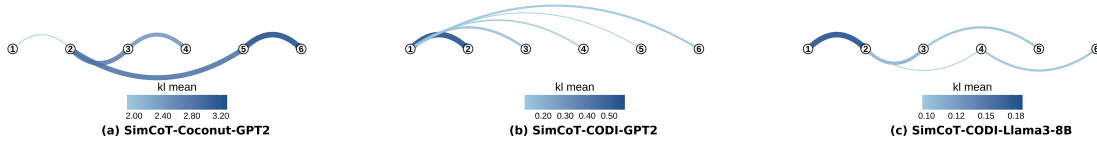


Figure 20. **RQ2 influence graphs for SIM-CoT on GSM8K.** The same principal-graph rendering as Figure 6 is used. Sim-CoT shows more local connectivity than the strongest skip-heavy COCONUT/CODI cases, while still displaying non-uniform step-to-step influence.

## E. Additional Model Size Analysis

We further replicate the RQ1 and RQ2 analyses on additional COCONUT backbones, including Qwen3-1.7B, Mistral-7B, DeepSeek-R1-Qwen2.5-1.5B, and Llama3-8B. These experiments are intended as a qualitative model-size/backbone check rather than a full scaling-law study, because the backbones differ in pretraining, instruction tuning, and reproduction details.

Figure 21 shows that step-wise flip rates remain heterogeneous across the additional backbones. The absolute flip rate varies substantially by model, but the profiles are not uniformly flat over steps. This supports the main RQ1 claim that latent steps have unequal causal leverage.

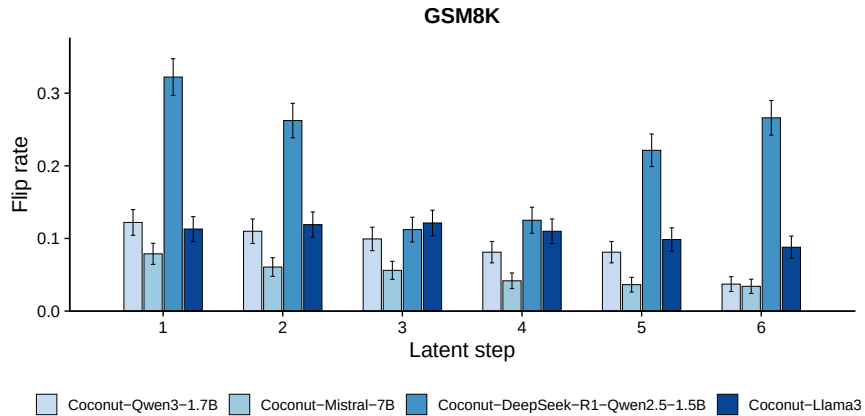


Figure 21. **RQ1 necessity check on additional COCONUT backbones (GSM8K).** The magnitude of decision flips differs across backbones, but step-wise sensitivity remains heterogeneous.

Figure 22 reports the corresponding RQ2 influence graphs. The additional models again show non-uniform propagation, with several long-range edges and model-specific routing patterns. Thus, the main influence-structure observation is not

confined to GPT-2/Llama3-1B/Qwen3-4B-Instruct settings, although a systematic scaling study over matched model families remains future work.

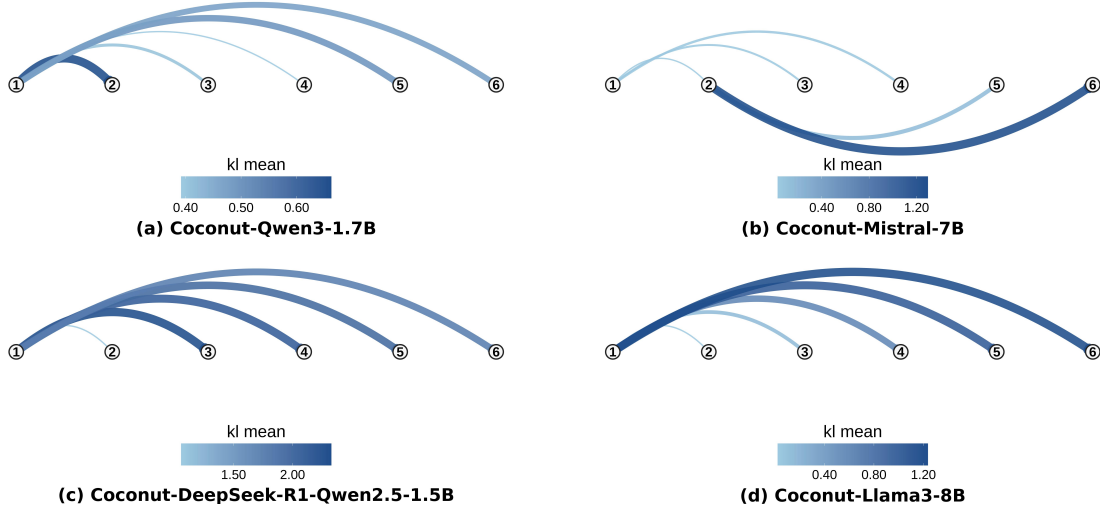


Figure 22. **RQ2 influence graphs on additional COCONUT backbones (GSM8K).** The graphs show non-uniform propagation and model-specific long-range routing across larger and alternative backbones.

## F. Additional RQ3 Details

### F.1. Trajectory sampling and latent-state collection

RQ3 studies multi-mode latent dynamics by analyzing multiple trajectories per input. For each example  $x$ , we generate  $N$  independent rollouts under the same decoding configuration as in the main experiments (temperature/top- $p$ /top- $k$  when sampling; greedy decoding when deterministic rollouts are required). For each rollout, we record (i) the final decoded answer  $\hat{y}$  and (ii) the realized latent states  $h_{1:T} = (h_1, \dots, h_T)$  at the model-defined latent steps, where each  $h_t \in \mathbb{R}^d$  is the last-layer hidden state associated with step  $t$ .

**Categorizing answers and defining modes.** We categorize final answers into a normalized form (e.g., extracting the final option letter for multiple-choice tasks, or applying the same numeric/boolean normalization used for evaluation). Given the set of normalized answers from  $N$  rollouts for the same input, we define the two dominant modes  $A$  and  $B$  as the two most frequent terminal answers. Rollouts whose terminal answers fall outside  $\{A, B\}$  are treated as residual modes and are excluded from the binary-mode analysis for that input. This procedure yields a set of labeled latent trajectories  $\{(h_{1:T}^{(i)}, m^{(i)})\}_{i=1}^M$  with  $m^{(i)} \in \{A, B\}$ , where  $M \leq N$  after filtering.

### F.2. Intermediate-step readout implementation details

**Probes for step-wise mode tracking.** To measure how mode information evolves across latent steps, we train lightweight probes on frozen latent states. For each step  $t$ , we fit a classifier  $\pi_{\phi,t}$  that maps the realized latent state  $h_t$  to a distribution over modes:

$$\pi_{\phi,t} : \mathbb{R}^d \rightarrow \Delta^2, \quad (\hat{s}_A(t), \hat{s}_B(t)) = \pi_{\phi,t}(h_t), \quad (20)$$

where  $\hat{s}_A(t)$  and  $\hat{s}_B(t)$  denote the probe-predicted probabilities of modes  $A$  and  $B$  at step  $t$ . Unless otherwise stated, we use linear probes (logistic regression) trained with cross-entropy loss and standard  $\ell_2$  regularization, which keeps probe capacity minimal and reduces the risk of overfitting artifacts unrelated to the model’s latent dynamics. When class imbalance arises after mode filtering, we balance the probe training set by subsampling to equalize the number of trajectories per mode.

**Teacher-forced readouts used in RQ3.** In addition to probe-based tracking, we use teacher-forced readouts to deterministically score candidate answers and reduce sampling noise when needed. We follow the same teacher-forcing

protocol described in Appendix A.3 (Teacher-forced readouts). Importantly, the canonical answer template is method-dependent rather than dataset-dependent: for COCONUT we use “[prefix] ### {answer}”, while for CODI we use “[prefix] The answer is {answer}”. Given a readout step  $t$ , teacher forcing provides a deterministic score (e.g., token-aggregated log-probability) for each candidate answer string, which we use as a complementary signal to the probe outputs in robustness checks.

### F.3. Supporting RQ3 analysis on GSM8K

The main RQ3 analysis uses StrategyQA because its binary answer space provides a clean setting for comparing two answer modes. We additionally run a supporting analysis on GSM8K by retaining prompts for which stochastic rollouts yield two sufficiently frequent dominant numeric answers. Because GSM8K has an open-ended answer space, this filtering is stricter and the retained set is smaller; we therefore use the result as supporting evidence rather than as the central RQ3 experiment.

Figure 23 shows that the probe-based score remains relatively stable through early and middle latent steps and then drops at the final step, while the teacher-forced log-probability readout is more model-dependent. This is broadly consistent with the main RQ3 interpretation: output-level answer preference and representational commitment need not collapse at the same step, but the GSM8K result should be read cautiously because of the harder two-mode filtering.

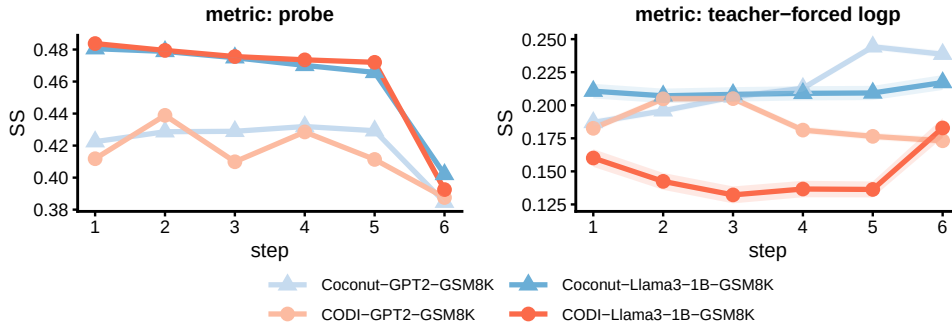


Figure 23. **Supporting RQ3 analysis on GSM8K.** Probe-based and teacher-forced readouts are computed over prompts with two dominant sampled numeric answers. The result supports the distinction between output-level preference and representational commitment, but is less central than StrategyQA due to GSM8K’s open-ended answer space.